Fiscal Stimulus and Household Debt Through the Lens of 80-million Loan Accounts*

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September 11, 2016

Abstract: The Thai government has implemented several measures to stimulate the economy in the past decade. These measures have impacted not only economic growth but they also affected financial situation of households and consequently the country's financial stability. However, our knowledge on the impacts of these measures is still limited. One of the reasons is due to the lack of comprehensive micro data that allow us to analyze financial behaviors of households and firms. This study provides a new perspective for the analysis of government measures and household debt, using account-level data from the National Credit Bureau (NCB). Since the NCB data spanned over the period that covered various significant events in the Thai economy, the data can be used to analyze changes in household debt in response to macroeconomic conditions. In addition, the data allow us to study the geographic distribution and accumulation of household debt in Thailand. Finally, this study illustrates an application of the NCB data to public policy analysis, investigating the short- and medium-term impacts of government stimulus through the first-car buyer tax rebate program on household debt. The findings from this study help policymakers better understand household finance in Thailand and design stimulus programs in the future.

* This paper is prepared for the 2016 Bank of Thailand Symposium. Chantarat and Lamsam: Puey Ungphakorn Institute for Economic Research, Bank of Thailand; Muthitacharoen: Faculty of Economics, Chulalongkorn University, Thailand; Samphantharak: School of Global Policy and Strategy, University of California San Diego. We would like to thank Thailand's National Credit Bureau (NCB) for providing us with the data used in this study as well as useful advice. Sirirat Sonjai, Bhumjai Tangsawasdirat and Chonnakan Tittinon provide excellent research assistance. The views expressed in this study are our own and do not represent those of the Bank of Thailand or the National Credit Bureau.

This study consists of two parts. **Part I** provides a discussion on the background of Thailand's National Credit Bureau (NCB) and its data. It then presents selected stylized facts about household debt in Thailand based on the anonymized statistical data provided by the NCB. **Part II** of the paper illustrates how administrative micro-level data could be used to analyze the impacts of government programs on household debt that ultimately help inform policymakers on the design of future fiscal stimulus programs.

The views expressed in this study are the researchers' and do not represent those of the Bank of Thailand or the National Credit Bureau.

"ข้อสงวน : บทความ บทวิเคราะห์ หรือบทวิจัยนี้ จัดทำขึ้นโดยสถาบันวิจัยเศรษฐกิจป๋วย อึ๊งภากรณ์ และบริษัท ข้อมูลเครดิตแห่งชาติ จำกัด เพื่อประโยชน์ในการเผยแพร่ต่อสาธารณะ ข้อมูลที่ปรากฏในรายงานฉบับนี้ จัดทำโดยอาศัยข้อมูลเชิงสถิติข้อมูลเศรษฐกิจการเงินจากแหล่งข้อมูลที่เชื่อถือได้ อย่างไรก็ตาม สถาบันวิจัยเศรษฐกิจป๋วย อึ๊งภากรณ์ และบริษัท ข้อมูลเครดิตแห่งชาติ จำกัด ไม่ยืนยันและไม่รับรองถึงความครบถ้วนสมบูรณ์หรือถูกต้องของข้อมูล จึงไม่รับผิดชอบต่อการนำเอาข้อมูล ข้อความ ความเห็น หรือบทสรุปที่ปรากฏในรายงานฉบับนี้ไปใช้ ไม่ว่ากรณีใดๆ สถาบันวิจัยเศรษฐกิจป๋วย อึ๊งภากรณ์ และบริษัท ข้อมูลเครดิตแห่งชาติ จำกัด มีสิทธิแต่เพียงผู้เดียวในทรัพย์สินทางปัญญาของรายงานฉบับนี้ และขอสงวนลิขสิทธิ์ในข้อมูลที่ปรากฏในเอกสารนี้ ห้ามมิให้ผู้ใดใช้ประโยชน์ทำซ้ำดัดแปลงนำออกแสดง ทำให้ปรากฏหรือเผยแพร่ต่อสาธารณชนไม่ว่าด้วยประการใดๆ ซึ่งข้อมูลในเอกสารนี้ไม่ว่าทั้งหมดหรือบางส่วนในเชิงพาณิชย์ เว้นแต่ได้รับอนุญาตเป็นหนังสือจากสถาบันวิจัยเศรษฐกิจป๋วย อึ๊งภากรณ์ และบริษัท ข้อมูลเครดิตแห่งชาติ จำกัด เป็นการล่วงหน้า นอกจากนี้ การกล่าวคัด หรืออ้างอิงข้อมูลบางส่วนตามสมควรในรายงานฉบับนี้ ไม่ว่าในบทความ บทวิเคราะห์ บทวิจัย ในเอกสารหรือการสื่อสารอื่นใด จะต้องกระทำโดยถูกต้องและไม่เป็นการก่อให้เกิดการเข้าใจผิด หรือความเสียหายแก่สถาบันวิจัยเศรษฐกิจป๋วย อึ๊งภากรณ์ และบริษัท ข้อมูลเครดิตแห่งชาติ จำกัด รวมทั้งต้องรับรู้ถึงความเป็นเจ้าของลิขสิทธิ์ในข้อมูลของสถาบันวิจัยเศรษฐกิจป๋วย อึ๊งภากรณ์ และบริษัท ข้อมูลเครดิตแห่งชาติ จำกัด และต้องอ้างอิงถึงฉบับที่และวันที่ในเอกสารฉบับนี้ของสถาบันวิจัยเศรษฐกิจป๋วย อึ๊งภากรณ์ และบริษัท ข้อมูลเครดิตแห่งชาติ จำกัด โดยชัดแจ้ง"

Part I

Household Debt Through the Lens of 80-million Loan Accounts

1. Introduction

1.1 Background on Thailand's National Credit Bureau

The concept to establish a credit data center started in 1961 and had been developed until 1995. The progress was interrupted by the 1997 financial crisis when many financial institutes were shut down. Following the crisis, the establishment of the credit data center was divided into two sections, one set up by the Government Housing Bank that established Thai Credit Bureau Co., Ltd. and the other one by the Bank of Thailand together with the Thai Banker's Association that established Central Information Services Co., Ltd. Both companies were founded with the same objective—to gather clients' data to help reduce the risk and increase the efficiency of giving loans and to prevent bad debt or non-performing loan problems in the economic system. In 2005, the two companies merged and became National Credit Bureau Co., Ltd.[1]

1.2 National Credit Bureau Data

Since the establishment of National Credit Bureau (NCB), there have been several financial institutions becoming members and submitting account-level credit information to the bureau. As of the first quarter of 2016, there were 89 members, including: (1) 24 banks (all 15 Thai commercial banks, four subsidiaries and branches of foreign banks, five special financial institutions—Government Savings Bank, Government Housing Bank, Export-Import Bank, SME Development Bank, and Islamic Bank) and (2) 65 non-bank financial service provider such as credit card, personal loan, and hire purchase leasing companies, as well as savings cooperatives). As a result, NCB maintains a rich database that covers 85% of the total credit to household and non-financial corporate sectors in Thailand, or 45% of the total financing in the financial system (including fund raising through debt and equity markets). The data cover 85% and 59% of the total

[1] The background on Thailand's National Credit Bureau is summarized from https://www.ncb.co.th/Company_Profile_en.htm

credit in Thailand's commercial banking system and special financial institutions (SFIs), respectively. Note that the data used in this study end in the first quarter of 2016 and do not include credit information from the Bank of Agriculture and Agricultural Cooperatives (BAAC), which began submitting its information to NCB in the second quarter of 2016, as well as additional savings cooperatives that continuously become NCB's members.

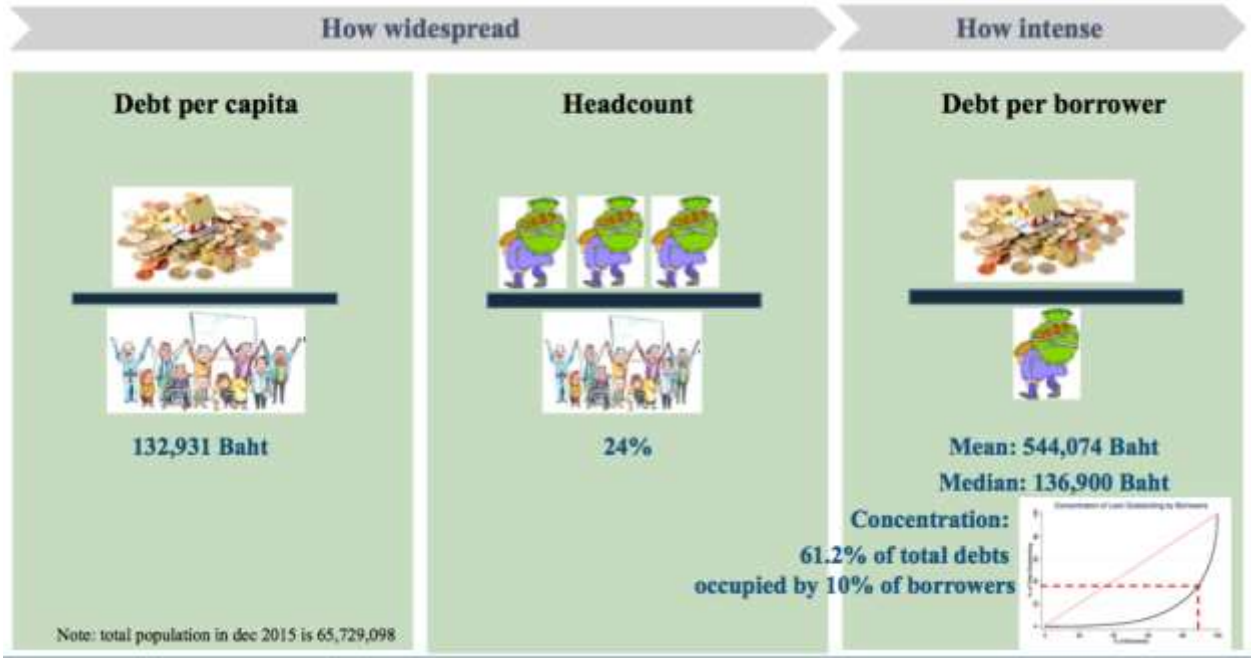
Credit information that members submit to NCB monthly consists of two components: information about each borrower and information about each loan account. Through the collaboration between NCB and Puey Ungphakorn Institute for Economic Research (PIER) at the Bank of Thailand, this study uses *anonymized statistical data* from NCB to study the development and current situation of debt held by households in Thailand. The data are special in several dimensions that help us better understand various aspects of the financial situation of Thai households. First, the data cover wide population. Specifically, as of the end of March 2016, the data account for the total loan outstanding of 8.7 trillion baht or 74% of the total household debt from 16.05 million borrowers and 49.1 million active accounts. The data also provide a consolidated view of financial institution relations for each borrower. Second, the data contain account level information that allows the analysis of household debt at a granular level that could unveil the heterogeneity across households. This is in turn highly important and relevant for policymaking.

2. A New Perspective of Household Debt Through the Lens of Credit Bureau Data

In this study, we present selected stylized facts from the NCB data that shed new light on household debt in Thailand. The main outcomes focused in this part are loan outstanding and delinquency. For each outcome, we are interested in the distributional aspects of debt by location and age.

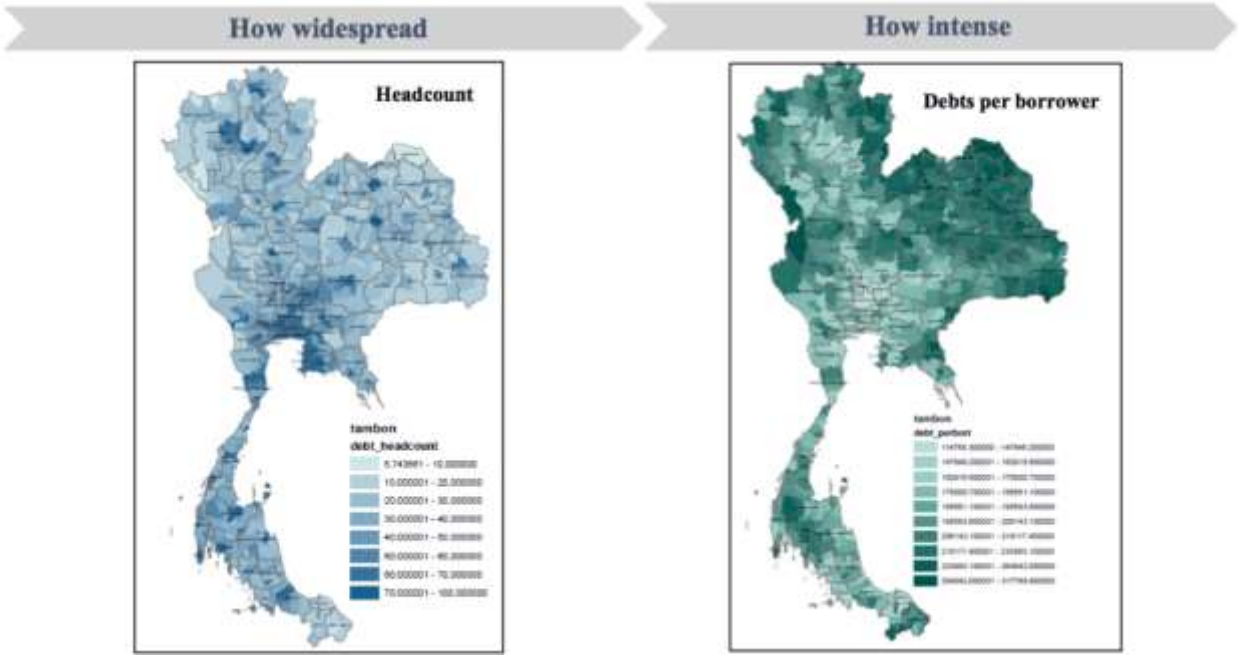
2.1 A Snapshot of Household Debt, March 2016

- Only 24% of population borrow
- Debt burden is highly concentrated



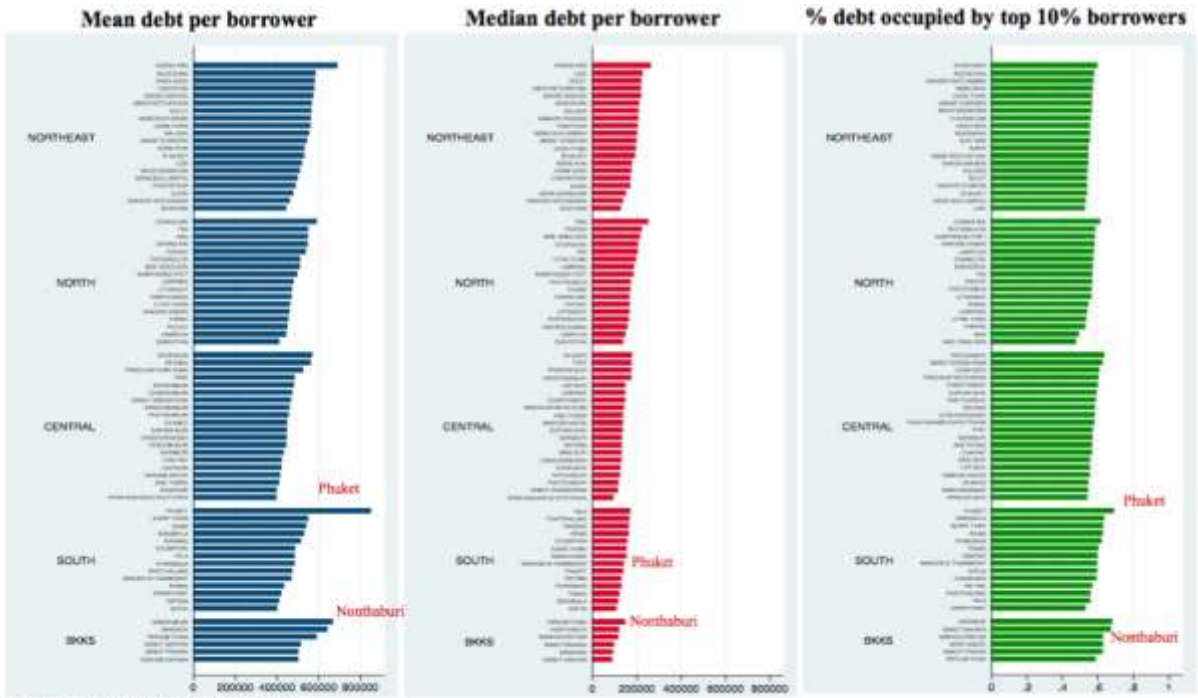
Geographic Distribution of Household Debt

- Distribution of debt intensity, defined as debt per borrower, is very different from that of headcount
- There is high debt intensity for borrower in the Northeast



Provincial Distribution of Debt per Borrower

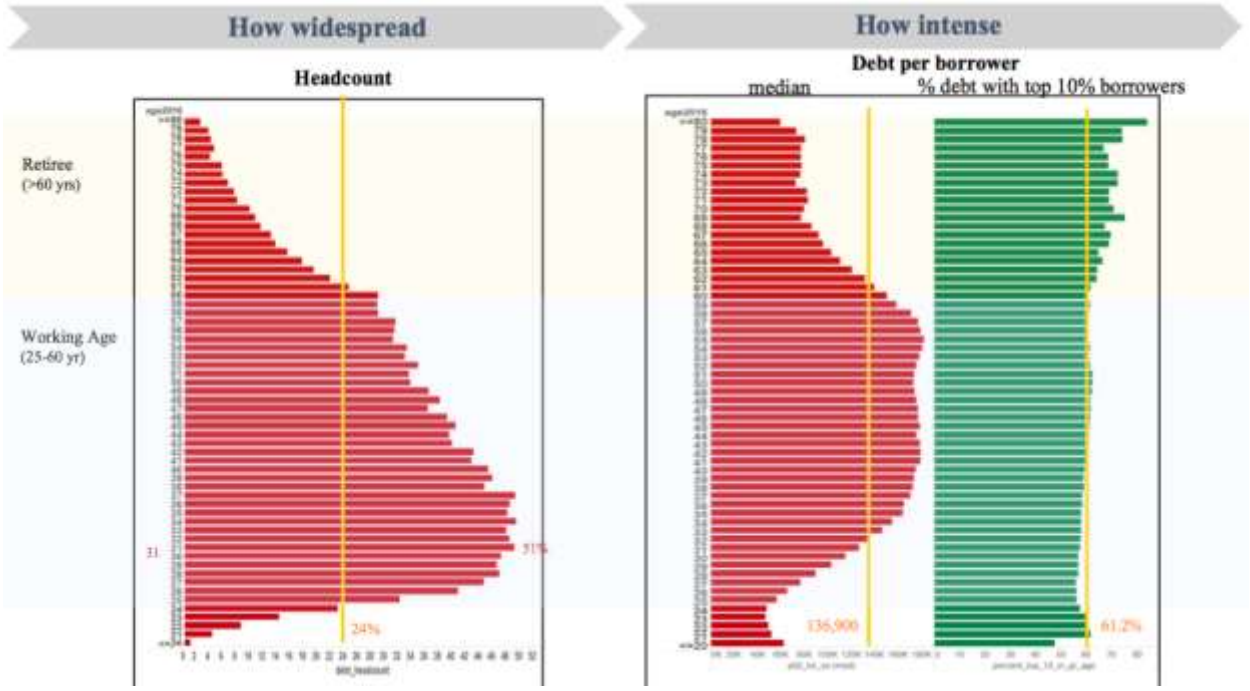
- Concentration is high across regions, especially Bangkok and vicinity
- Median debt is also relatively high for borrowers and the Northeast



Note: Central region includes East and West region.

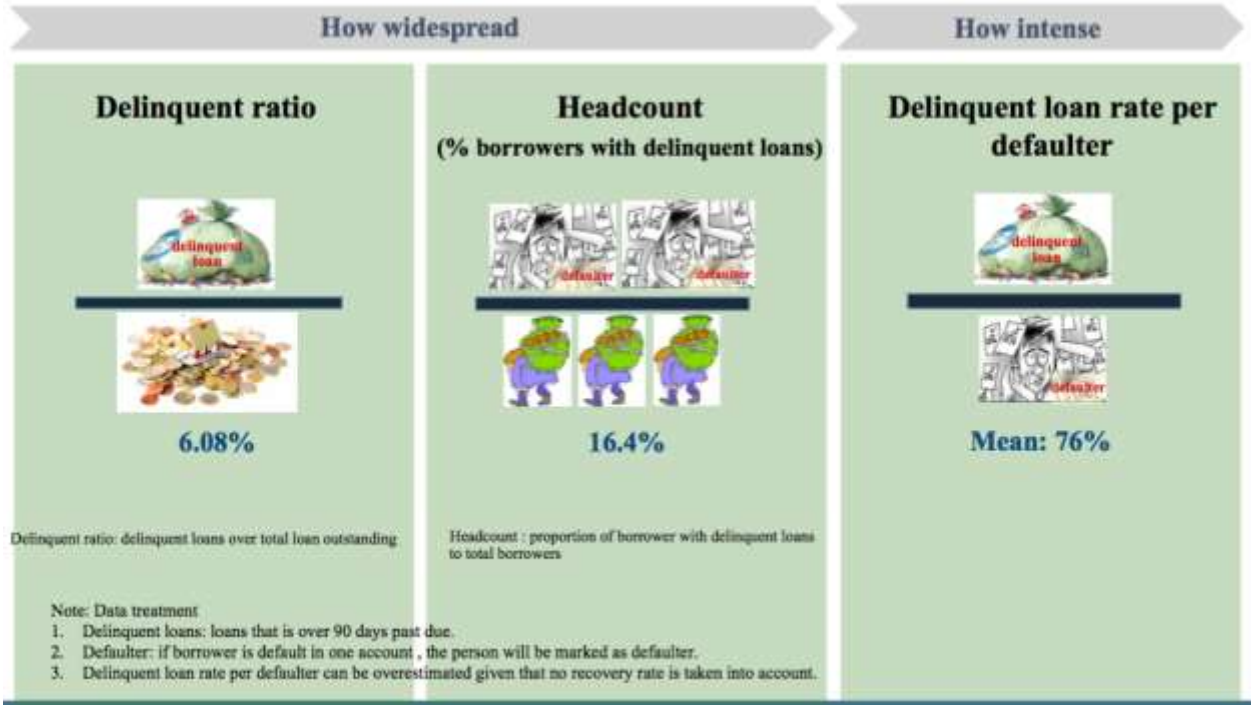
Household Debt Distribution by Age Group

- Debt burden follows life-cycle pattern with increasing debt concentration for retirees
- Several questions arise: Too soon to have debt among the young? Too long debt duration over the life cycle? Too much debt held by the old?



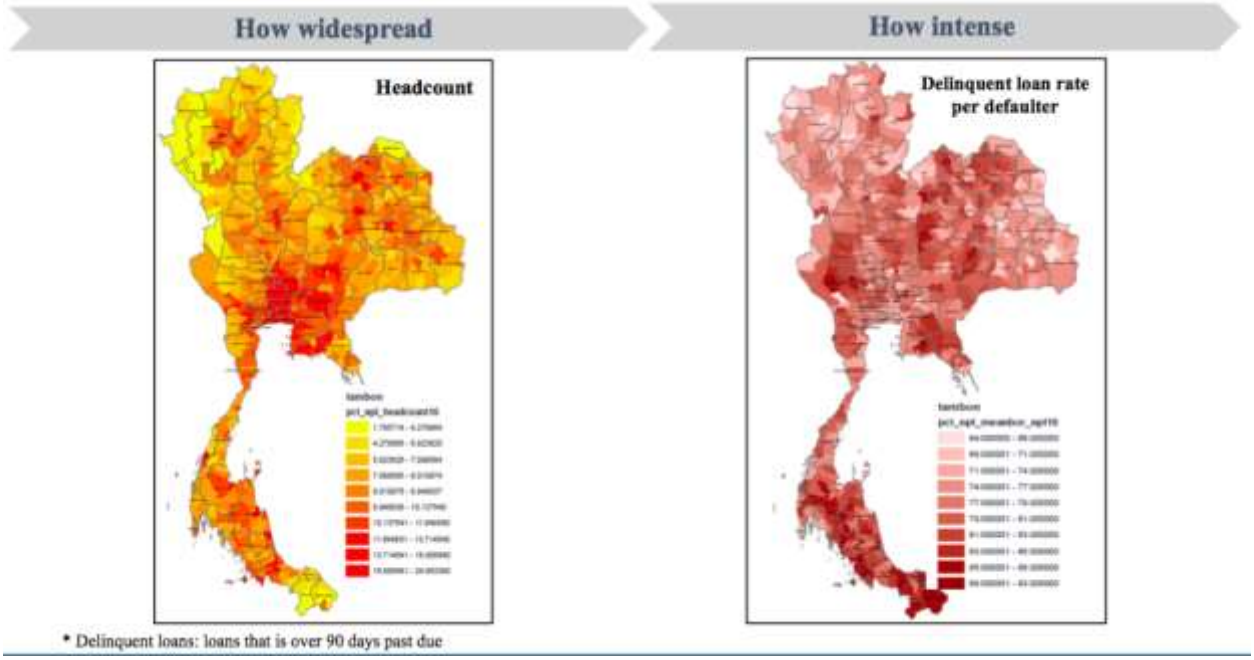
2.2 A Snapshot of Delinquent Loan, March 2016

- 16.4% of borrowers have delinquent loans; for those that do, delinquency ratio is high



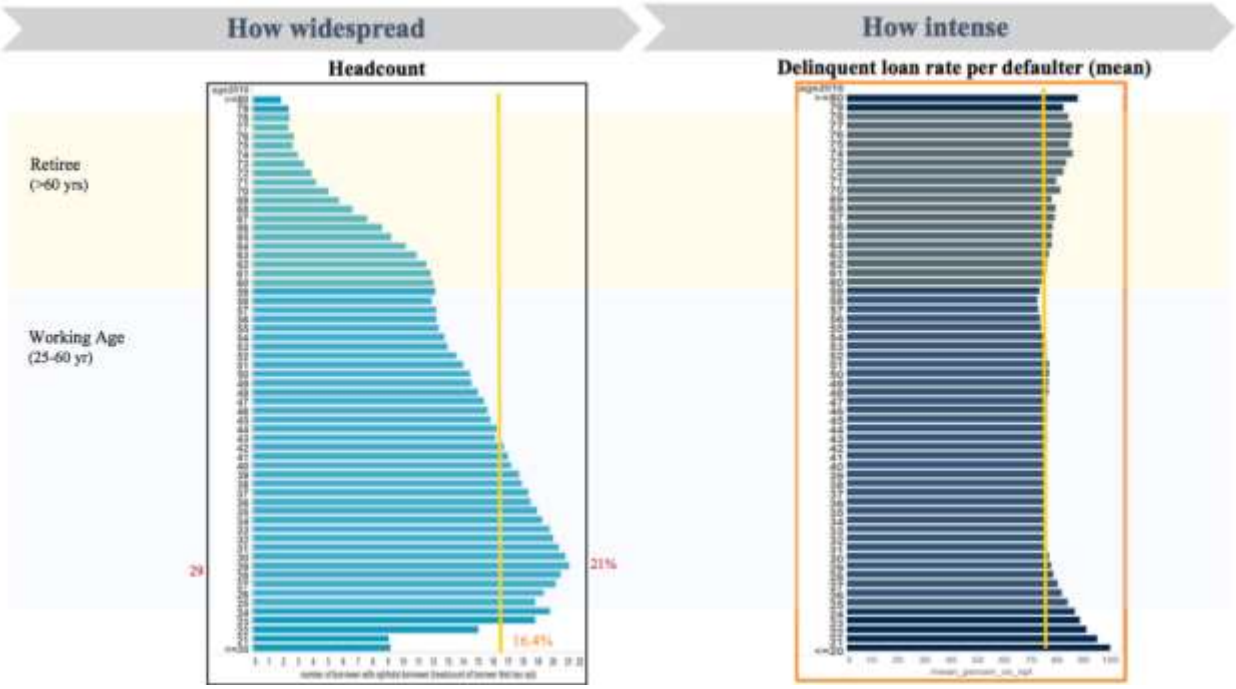
Geographic Distribution of Delinquent Loan

- Delinquent loan headcount is relatively high for Bangkok and vicinity as well as the Northeast



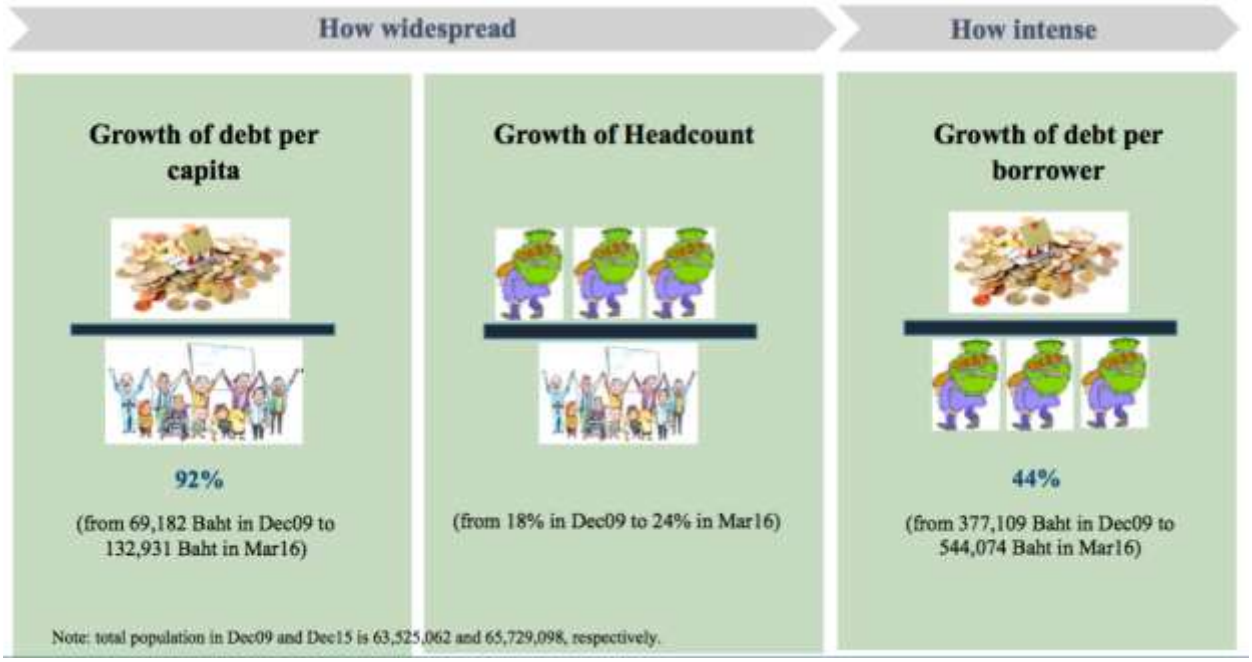
Distribution of Delinquent Loan by Age

- Delinquent loan headcount is high for young adults in their early 30s



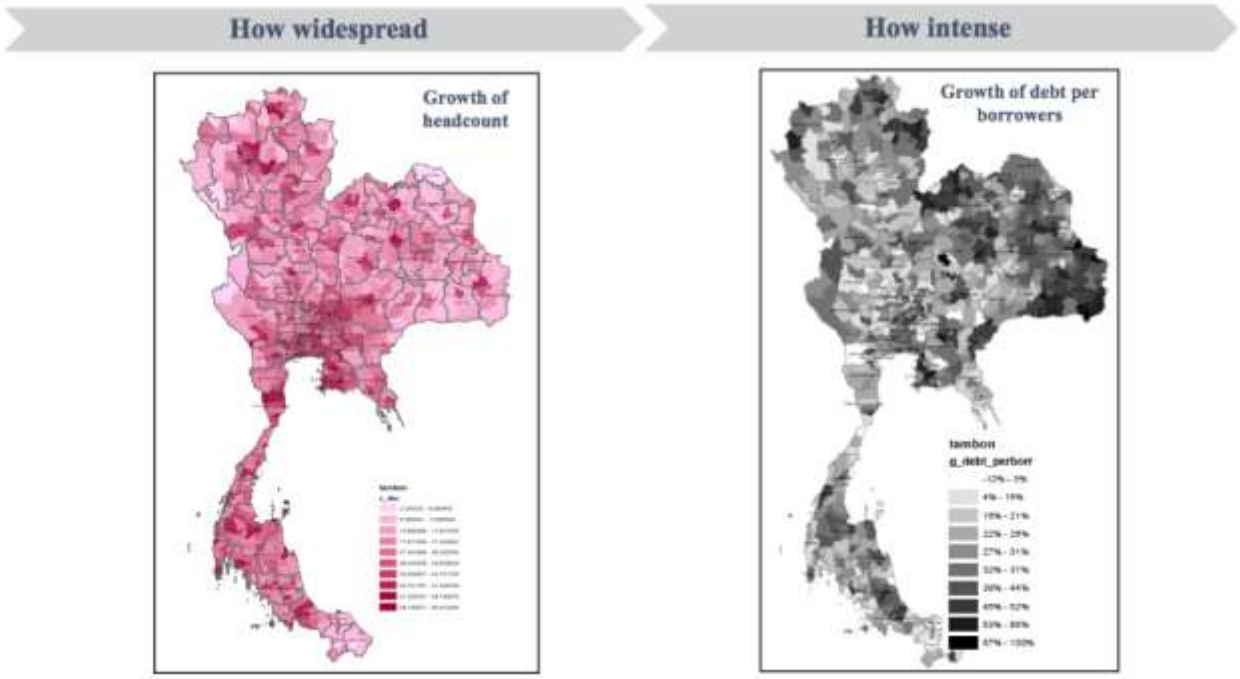
2.3 Debt Growth, 2009-2016

- Debt growth from 2009-2016 comes from both increasing headcount as well as increasing amount of debt per borrower



Geographic Distribution of Debt Growth

- There is catch-up growth in credit in the Northeast



Part II

An Application: Using National Credit Bureau Data to Analyze Unintended Effects of Durable-Goods Stimulus

1. Introduction

Durable goods subsidies have been used by various governments as a stimulus to generate domestic demand during economic slowdown. One of the key advantages of this policy is that it can bring forward aggregate demand from the future, where the economy is operating at close to its capacity, to the present, where there is substantial slack. However, its opponents often contend that private sector offsets such effects of fiscal intervention almost immediately, negating any positive effects. In addition, durable-goods stimulus could also produce important—many unintended—effects on individuals who borrow to finance their durable-goods purchases. If it leaves adverse impacts on the program participants, it could undermine potential benefits of the stimulus on the economy. In this paper, we aim to study the impact of the program on one aspect of individuals who participate in the stimulus, namely, their debt-servicing capacity as well as propensity to acquire new loan. Specifically, we ask how the durable-goods stimulus affects loan portfolio and loan performance of individuals. In order to answer this question, we use unique account-level loan data from Thailand's National Credit Bureau to study the impact of the first-car buyer tax rebate scheme that was implemented in the country during 2011-2012.

In particular, we use difference-in-difference estimation to study the impacts of the first-car tax rebate program, where the treatment group consists of first-time buyers that purchased qualified car models during the period of the program while the control group is comprised of first-time buyers that purchased the same car models but outside the program period. We focus our analysis on two sets of outcome variables: delinquency probability and new loan origination probability. First, we find higher delinquency probability for individuals in the treatment group relative to the control group. Our results also show that the rising delinquency probability was not limited to only the first car, but there was an evidence of higher delinquency on other loans for those in the treatment group as well. We find that the impacts of the program on delinquency probability vary across different types of buyers. The impact on delinquency was lower for those with prior

commercial bank access. Second, our results suggest that new loan origination probability of the individuals in the treatment group was significantly lower relative to that of the control group. Our study also shows that the negative impact of the program on new loan origination is smaller for those buyers who had prior access to commercial banks, older buyers, and buyers living in the urban areas.

In summary, our results suggest that, as the results of the first-car buyer tax rebate program, buyers with prior access to commercial banks, older buyers, and buyers who live in the urban areas tended to experience lower delinquency probability and have higher new loan origination probability. Intuitively, given that these buyers generally had more financial access prior to the program, participating in the program and pulling demand from the future did not much jeopardize their ability to take loans as relative to the buyers without prior access commercial banks, younger buyers, and rural buyers, whose first-car loans came at a cost of reducing other loans.

The rest of this paper is as follows. Section 2 describes background on the first-car buyer tax rebate scheme. Section 3 focuses on the analysis of the program impacts at the individual level. Section 4 turns to the analysis of the program at the local economy level. Section 5 concludes.

2. Background on Thailand's First-Car Buyer Tax Rebate Program

The first-car buyer tax rebate program in Thailand was an excise tax rebate scheme for first-time car buyers. The program was first introduced to the public by Pheu Thai Party in June 2011 during its campaign for the July 2011 general election. Following the election, the government led by Pheu Thai Party made the program official in August 2011. The program's objectives declared by the government were: (1) to encourage car ownership, (2) to stimulate economic growth through supports for the automobile-related industries, and (3) to increase the government's tax revenues.

The program consisted of an excise tax rebate of 50,000-100,000 baht depending on the car models. There were several qualifications that each car buyer must satisfy in order to participate in the program. First, a participant must be a first-time car buyer. Second, the buyer must purchase one of the qualified models of passenger cars or trucks. The qualified models were in turn declared

by the government. For passenger cars, the qualified models were those under the price limit of 1 million baht and with engines not larger than 1,500 CC. For trucks, the price limit was also 1 million baht but there was no restriction on the engine size. Third, the car must be purchased or reserved during September 2011 – December 2012. Finally, the buyer must possess the car for at least five years.

The qualified participant received an excise tax rebate about one year after possessing the car. It is important to note that the government allowed participants to just submit the reservation receipt before the end of 2012. This allowed the program participants to make the actual purchase, and thus originate the car loan, after 2012. In addition, the severe flood that hit the majority of the country in the second half of 2011 caused many interruptions in domestic car production and resulted in a large number of participants receiving their cars in 2013. In this study, we therefore assume that the program period is from September 2011 to December 2013. According to the Excise Department, more than 90 percent of the participants received their cars by the end of 2013. In total, about 1.25 million individuals participated in the program. To get a sense of the size, the number of total car purchases under the Cash-for-Clunker program in the US in 2009 was just about 680,000. This suggests that the first-car buyer rebate program likely produced noticeable impacts on the Thai economy.

The first car program shares key features with other durable-goods stimulus that has been implemented in other countries. For example, France implemented subsidies for scrapping an old car with a new one during 1994-1995. Following the Global Financial Crisis, several countries implemented similar programs to revive auto demand. Although the details of these programs varied, most had the aim to remove inefficient vehicles from the road. Examples include the Cash-for-Clunkers program in the US in 2009 as well as similar programs in France in 2009 and Japan in 2009-10. These programs involved offering an incentive for individuals to purchase durable goods in a given time frame, effectively moving future demand to the present. However, unlike previous studies that examined only general equilibrium effects of such stimulus on the local economies[2], the first-car rebate program provided us an opportunity to study the micro (individual-

[2] For example, see Mian and Sufi (2010).

and loan-level) effects of durable goods stimulus on those who were offered monetary incentives to participate. Combining with the unique credit bureau dataset, this stimulus program offers a unique opportunity to examine the effects on individuals who finance the purchases with loans.

3. Empirical Analysis at the Individual Level

3.1 Data

The data in this study are account-level loan data from Thailand's National Credit Bureau (NBC). We begin with a stratified random sample that oversamples individuals with car loans and construct a quarterly panel of individuals from December 2009 to March 2016. The sample contains about 100,000 individuals with information on loan composition, loan history, debt payment, and basic demographics.

Since this paper focuses on the impacts of tax rebates on first car buyers, we define first-car purchases by identifying the first car loan for each individual in our sample that did not have any car loan prior to that origination.[3] In addition, given that the program was only applied to certain car models, we further limit our sample to include only first-time car buyers who purchased car models that satisfied the program's criteria. Specifically, for passenger cars the qualified models were those under the price limit of one million baht and with engines not larger than 1,500 CC. For trucks, the price limit was also one million baht but there was no restriction on the engine size. Our final sample contains 18,829 individuals (169,271 observations). About half of the sample are buyers of passenger cars.

Lastly, we divide the final sample into treatment and control groups. The treatment group is defined as those who originated loans during September 2011-December 2013 since they were likely to participate in the first-car tax rebate program. In contrast, the control group consists of first-time buyers who originated loans outside the above-mentioned period and were not eligible

[3] Given our strategy, our sample misses first-time buyers who purchased car in cash (i.e. no loan origination) and includes non-first car buyers whose earlier purchases were either in cash or with loans originated before December 2009.

for the program. The treatment group accounts for approximately 70% of the sample. Note that there are several candidates for control groups. One possibility could be individuals who purchased their first car during the program period, but the car models were not qualified for the tax rebate. Another candidate is individuals who purchased qualified car models during the program period, but were not qualified for the rebate because the purchases were not their first time. None of these candidates, including the one we use in this paper, is a perfect control group because the decision to purchase the cars was not random, implying that individuals in the treatment and the control groups possessed some different characteristics. The rationale behind our decision to choose individuals who were first-car buyers that purchased qualified car models outside the program period as our control group is that we would like to focus on the effects of moving forward future demand to the present. Our control individuals were first-time buyers that purchased the same car models as the treatment counterparts. The major difference from the treatment individuals is that they did not move forward their demand to the present.

Table 1: Summary Statistics

Variables	Control			Treatment			Treatment-Control
	Mean	SD	n	Mean	SD	n	
Demographics							
Age	42.4	9.8	5,162	40.1	10.1	13,667	-2.36***
							(0.16)
Urban	0.5	0.5	5,162	0.5	0.5	13,667	-0.01
							(0.18)
Portfolio							
Number of loan accounts	8.7	9.8	2,528	8.6	9.5	5,830	-0.15
							(0.23)
Having a mortgage	0.2	0.4	2,528	0.2	0.4	5,830	-0.02*
							(0.01)
Having a personal loan	0.6	0.5	2,528	0.6	0.5	5,830	-0.01
							(0.01)
Having credit card(s)	0.4	0.5	2,528	0.4	0.5	5,830	-0.07***
							(0.01)
Having commercial bank loans	0.6	0.5	2,528	0.5	0.5	5,830	-0.06***
							(0.01)

Notes: Numbers in parentheses indicate standard error. *** = Significantly different from zero at the 1% level, ** = Significantly different from zero at the 5% level, * = Significantly different from zero at the 10% level.

Table 1 provides summary statistics of demographic and portfolio characteristics for the treatment and the control groups in our sample. Individuals in the treatment groups, on average, are about 40 years old—roughly only two years younger than those in the control group. The degree of urbanization is about 50% for both groups. Individuals in the two groups also have quite similar portfolio characteristics. The differences between these two groups in some dimensions are statistically significant but the magnitudes are all very small.

3.2 Empirical Strategy

3.2.1 The Basic Specification

We use difference-in-difference strategy to estimate the impacts from the first-car buyer tax rebate program. However, given that individuals in our treatment and control groups purchased their cars in different periods (quarters), we make an adjustment by first defining the purchase quarter as period zero. The quarters prior to the purchase are then defined as periods -1, -2, and -3 while the quarters following the purchase are referred to as periods 1, 2, 3, and so on. The rationale for including the quarters before the purchase into our analysis is that the program implementation was not a surprise and there could be an impact of the program on debt portfolios and performance even before the car purchase due to individual's preparation for the transaction. Likewise, we include the quarters following the purchase to allow for the effects that could take longer than one quarter. Empirically, we include 10 quarters in our analysis: three quarters prior to the purchase, the purchase quarter itself, and six quarters following the purchase. With these lag and lead variables, our sample includes individuals who bought qualified passenger car or truck models as their first vehicle during the period 2010:Q1-2014:Q4.

Specifically, for the basic specification we estimate the following regression:

$$y_{it} = \sum_{m=-4}^{6} \alpha_m \, r_{i,m} + \sum_{m=-4}^{6} \beta_m \, treatment_i \cdot r_{i,m} + province_year\ FE + time\ FE + individual\ FE + \varepsilon_{it}$$

where y_{it} is an outcome variable; $r_{i,m}$ is an indicator variable for each lag and lead quarters, *m*; and $treatment_i$ is an indicator variable that equals 1 if individual *i* is in a treatment group (i.e., participants in the first-car scheme) and 0 otherwise. Again, the treatment group includes participants in the first-car scheme, i.e., first-time buyers who bought qualified passenger car or truck models during the program period (September 2011 - December 2013). The control group includes those who bought the same qualified models but outside the program period (January 2010 - August 2011 and January 2014 - December 2014). We also include time fixed effects to control for macroeconomic conditions that could impact our outcome variables and individual fixed effects to capture unobserved time-invariant characteristics of each individual.

3.2.2 Outcome Variables

We examine two sets of outcome variables in this study: (1) probability of loan delinquency for each individual; and (2) probability that each individual originated new loans besides the first car loan.

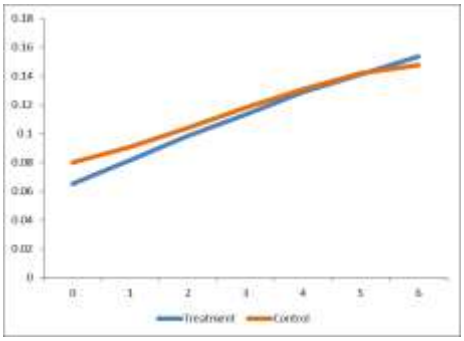
The first set includes the probability of loan *delinquency* for each individual. On the one hand, participants in the first car scheme received tax rebates from the government and the permanent income hypothesis would suggest that wealth of these individuals increased while that of the control group remained unaffected. Alternatively speaking, one may interpret that the rebates from the program lowered the effective amount of the new auto loan for the treatment group while the amount for the control group was unchanged. In this case, we should expect individuals in the treatment group to have lower delinquency than those in the control group. On the other hand, the first-car buyer tax rebate program likely encouraged individuals to move their future demand for (first) car to earlier periods. If these buyers were still not ready to service the newly originated loans, they could face higher delinquency on this new auto loan and/or other existing loans in their portfolios, relative to individuals in the control group whose first car purchase decisions were not distorted by the stimulus, i.e., they were more likely to originate new auto loans when they were more financially ready. Therefore, the net effect of the program on delinquency becomes an empirical question. Note that the effects on delinquency may not only limit to the buyers' first-car loans, but there could be spillovers to other loans in their loan portfolios as well.

The second set of outcome variables includes the probability that each individual originated other *new loans* besides the first car loan itself. The rationale behind our interest in these variables is that the new auto loans from the first car purchases could crowd in or crowd out other potential new loan originations. Generally speaking, while new auto loans could push the car buyers to their debt limits and therefore reduce their ability to acquire other loans, if the new cars were used in productive activities it is possible that the new cars could help individuals get other loans. This could be either from individual's higher income generated by car-related production activities that allowed them to take more loans, or from additional loans necessary for the operation of new businesses. For example, if a car was used by an individual to open a new business enterprise, this business may open an overdraft account or take personal loans to further finance its operation. The impacts of new auto loans on the probability of originating other new loans could therefore be different between the treatment and the control groups. On the one hand, the first-car buyer tax rebate program induced certain individuals to originate auto loan prematurely. Therefore, the program would likely push these individuals above their debt limit sooner, leading to a larger drop in the origination probability of other loans. On the other hand, it is also possible that the tax rebates from the stimulus helped facilitate entrepreneurship which in turn resulted in more new loan originations.

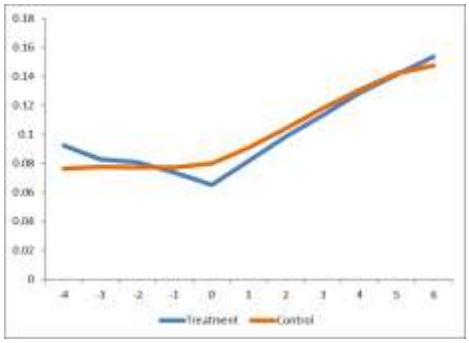
Finally, given that passenger cars are likely to be used mainly for consumption purpose while trucks are more likely to be used in business operations, we also analyze the impacts of the first car program on the outcome variables for passenger cars and trucks separately. Furthermore, to shed light on the adjustment of individual's portfolio choices in response to the program, for each set of outcomes we also separately examine the impacts on (1) all loans, (2) unsecure loans, i.e., credit cards, personal loans, and overdraft, and (3) other secure loans (excluding the first-car loan itself), i.e., mortgage and high purchase loans.

Figure 1: Means of outcome variables during the pre- and post-program period

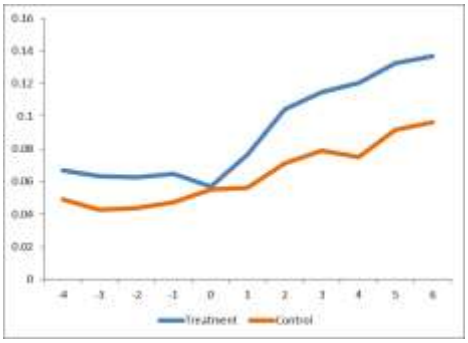
A: Delinquency (First car)



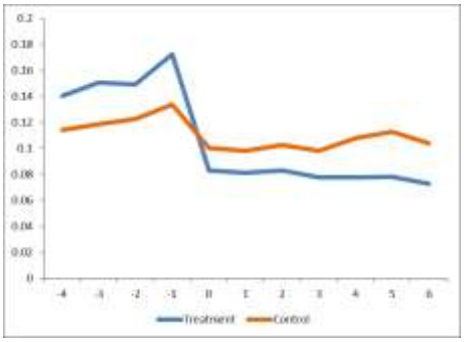
B: Delinquency (Unsecure)



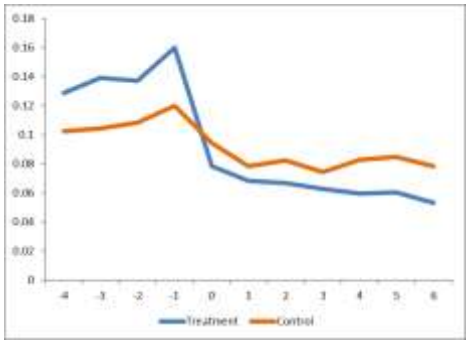
C: Delinquency (Secure)



D: New loan (All)



E: New loan (Unsecure)



F: New loan (Secure)

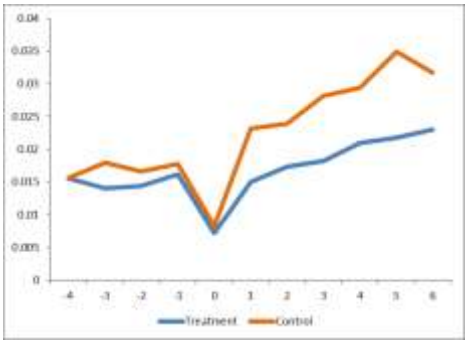


Figure 1 illustrates the pattern of the outcome variables before and after the car-purchase decision. It allows us to examine the common trend assumption for the treatment and the control groups. For the new loan variables (Panels D to F), the propensity to acquire new loans for the two groups moves almost parallel during the pre-purchase period. This is consistent with the common trend assumption before the program participation. The patterns for delinquency, however, are not fully parallel during the whole pre-purchase period (Panels A to C), calling for the need to control for the slope difference in the regression analysis.

Table 2: Basic Difference-in-Difference Framework (New Loan Propensity)

Panel A: New loan excluding first car (All)

	Pre	Post	Difference (Post-Pre)
Control	0.136***	0.125***	-0.011***
	(0.003)	(0.002)	(0.004)
Treatment	0.174***	0.097***	-0.077***
	(0.003)	(0.001)	(0.002)
Difference (Treatment-Control)	0.038***	-0.029***	-0.066***
	(0.004)	(0.002)	(0.004)

Panel B New loan excluding first car (Unsecure loans)

	Pre	Post	Difference (Post-Pre)
Control	0.121***	0.103***	-0.018***
	(0.003)	(0.002)	(0.004)
Treatment	0.161***	0.083***	-0.078***
	(0.003)	(0.001)	(0.002)
Difference (Treatment-Control)	0.040***	-0.020***	-0.060***
	(0.004)	(0.002)	(0.004)

Panel C: New loan excluding first car (Secure loans)

	Pre	Post	Difference (Post-Pre)
Control	0.019***	0.028***	0.009***
	(0.001)	(0.001)	(0.002)
Treatment	0.016***	0.018***	0.001
	(0.001)	(0.000)	(0.001)
Difference (Treatment-Control)	-0.003*	-0.010***	-0.008***
	(0.002)	(0.001)	(0.002)

Notes: Numbers in parentheses indicate standard error. *** = Significantly different from zero at the 1% level, ** = Significantly different from zero at the 5% level, * = Significantly different from zero at the 10% level

Table 2 shows the means of one of the outcome variables (new loan propensity) and their differences.[4] This table is provided as an illustration of the identification strategy. Panel A, for example, shows the propensity of having new loans (apart from the first car). For both control and treatment groups, the propensity in the post-purchase period is lower than that before the purchase. That drop, however, is much sharper among the treatment group. The difference in these differences is strongly significant. Similar findings are observed in Panels B and C. This simple experiment supports our identifying assumption. In the next section, we examine this impact using a more formal framework.

3.2.3 Impact Heterogeneity

In addition to the overall impact estimation, we further investigate whether there were heterogeneous impacts across different types of individuals. Empirically, we first classify individuals into two groups for each criterion and construct an indicator variable, *G*, based on this classification. We then include three-way interaction terms that include the indicator for treatment group, the indicator for lag/lead quarters, and the indicator *G*.

Specifically, for the impact heterogeneity specification we estimate the following regression:

$$y_{it} = \sum_{m=-4}^{6} \alpha_m \, r_{i,m} + \sum_{m=-4}^{6} \beta_m \, treatment_i \cdot r_{i,m} + \sum_{m=-4}^{6} \gamma_m \, treatment_i \cdot G_i \cdot r_{i,m} + province_year\ FE + time\ FE + individual\ FE + \varepsilon_{it}$$

where the variables are previously defined. If the regression coefficients of these three-way interaction terms, γ_m, are statistically significant, the result implies that there was heterogeneity in the program impacts across different types of individuals.

[4] The basic difference-in-difference framework for the other outcome variables (delinquency propensity) are included in the appendix.

We explore three criteria that categorize individuals in our sample into groups. First, we stratify first-time car buyers in our sample into two groups based on their prior access to commercial bank loans, defined as having loans from commercial bank five quarters before purchasing their first cars. Those with prior access to commercial banks were more likely to have better access to finance and/or have lower default risk so we expect that the impacts on delinquency probability for this group would be lower than those without prior access. The first-car loans from the rebate program were also less likely to crowd out other loans of buyers with prior access to commercial banks. Second, we stratify our sample by age. Buyers in the younger group are those born in or after 1980, i.e., they were less than 32 years old when the program was implemented in 2011. Everything else equal, we expect that younger buyers were less financially secure than the older, and we would observe more adverse impacts of the program on the former group's loan performance and portfolio than that of the latter group. Finally, we classify the buyers in our sample by residential areas. Those located in the rural areas are assumed to have more difficult access to finance than those living in the urban (municipality) areas, and experienced worse impacts of the program on their loans.

3.3 Empirical Results

3.3.1 Delinquency Probability

We begin our analysis by estimating the overall impacts of the first-car buyer tax rebate program on delinquency probability. Table 3 presents selected estimated coefficients for delinquency probability regression. For delinquency on the first car loans, we find higher delinquency probability for individuals in the treatment group relative to the control group for both passenger cars and trucks, consistent to the argument that the program encouraged individuals to originate auto loans while they were still not financially ready. The higher delinquency was evident since the first quarter following the car purchase and remained strongly significant throughout the five subsequent quarters. Furthermore, our results show that the rising delinquency probability was not limited to only the first car, but there was an evidence of higher delinquency on other loans for those in the treatment group when compared with the controlled individuals, although the results are weaker and less systematic when we break the sample down into passenger cars and trucks.

For those who purchase passenger cars, the delinquency probability was higher on both unsecure and secure loans (for secure loans, the coefficients had similar magnitudes but were imprecisely estimated). For those who purchase trucks, the impact on delinquency probability was also higher on both unsecure and secure loans, with stronger impacts on secure loans. It is worth noting the small number of observations on the secure loan specifications. This is due to the relatively small number of individuals who had other secure loans apart from the first car loan itself.

Table 3: Delinquency probability

	All			Passenger Cars			Trucks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	First car	Unsecure	Secure ex. First car	First car	Unsecure	Secure ex. First car	First car	Unsecure	Secure ex. First car
Treatment x r(-3)		-0.001	-0.004		0.008	-0.004		-0.014*	-0.003
		(0.005)	(0.007)		(0.007)	(0.014)		(0.006)	(0.015)
Treatment x r(-2)		0.001	-0.001		0.004	-0.002		0.002	0.004
		(0.007)	(0.006)		(0.007)	(0.014)		(0.008)	(0.023)
Treatment x r(-1)		-0.000	0.002		0.003	0.003		-0.000	0.004
		(0.007)	(0.012)		(0.010)	(0.024)		(0.005)	(0.014)
Treatment x r(0)		0.001	-0.012		-0.003	-0.012		0.011*	-0.004
		(0.009)	(0.006)		(0.016)	(0.019)		(0.005)	(0.017)
Treatment x r(1)	0.017***	0.013**	0.016	0.015***	0.016	0.017	0.019***	0.014*	0.024
	(0.001)	(0.004)	(0.017)	(0.002)	(0.010)	(0.022)	(0.003)	(0.007)	(0.013)
Treatment x r(2)	0.041***	0.022**	0.032	0.039***	0.024*	0.022	0.045***	0.026	0.059**
	(0.005)	(0.009)	(0.017)	(0.004)	(0.010)	(0.021)	(0.007)	(0.017)	(0.017)
Treatment x r(3)	0.056***	0.025**	0.040**	0.048***	0.029	0.024	0.066***	0.022	0.078**
	(0.005)	(0.007)	(0.014)	(0.004)	(0.015)	(0.015)	(0.008)	(0.019)	(0.028)
Treatment x r(4)	0.068***	0.025	0.046*	0.058***	0.036	0.034	0.080***	0.013	0.079
	(0.005)	(0.017)	(0.021)	(0.006)	(0.026)	(0.025)	(0.009)	(0.015)	(0.040)
Treatment x r(5)	0.079***	0.025*	0.036*	0.065***	0.035*	0.026	0.095***	0.014	0.057
	(0.008)	(0.013)	(0.016)	(0.008)	(0.015)	(0.022)	(0.010)	(0.023)	(0.034)
Treatment x r(6)	0.088***	0.020	0.041	0.074***	0.029*	0.030	0.102***	0.006	0.063
	(0.009)	(0.012)	(0.023)	(0.010)	(0.014)	(0.036)	(0.012)	(0.021)	(0.036)
Observations	131,264	97,328	28,434	66,705	59,674	16,925	64,559	37,654	11,509
Number of observations	18,829	10,674	3,817	9,571	6,436	2,195	9,258	4,238	1,622

Notes: The table reports the marginal effects of the treatment group at the quarter (m) relative to the purchase. Each regression includes both time- and individual-fixed effects. Standard errors are heteroscedasticity-robust and are clustered on cohort groups. *** = Significantly different from zero at the 1% level, ** = Significantly different from zero at the 5% level, * = Significantly different from zero at the 10% level.

When we allow for impact heterogeneity across individuals, we find that the impacts of the program on delinquency probability vary across different types of buyers. First, for passenger car buyers, the impact on delinquency was lower for those with prior commercial bank access, with significant effects on first car and unsecure loans. For truck buyers, we also find that the program impact was lower delinquency for those with prior commercial bank access, with significant effects on secure loans. These results suggest that those without access to commercial bank loans in the absence of the program were likely borrowers with higher default risk, highlighting the role of commercial banks in evaluating risk of potential borrowers. Second, we find that for truck purchases the impact on delinquency probability of unsecure loans for younger buyers was higher than that of the older, suggesting the higher risk of younger buyers. Finally, we find no systematic results of location (urban versus rural) on the program impact on delinquency probability, although the difference in the program impacts was detected for secure loans, both for passenger cars and trucks. The regression results are presented in Tables B.1-B.3 in the appendix.

3.3.2 New Loans

Next, we study the impact of the first-car buyer tax rebate scheme on the probability that individuals would originate other new loans. The results in Table 4 show that the effects on probability of acquiring new loans were evidently different between first-time buyers of passenger cars and trucks. The overall results suggest that, relative to the control group, the probability that individuals in the treatment group would originated new loans following the first car purchase was significantly lower. The table further reveals that this finding was driven by passenger car purchases and there was no impact of the program among truck buyers. For passenger cars, the impacts of the program lasted for at least six quarters since the purchase and were on secure loans, although the magnitudes of the effects are similar between unsecure and secure loans. Given that passenger cars and trucks are likely used in different activities (i.e., mainly as household consumption versus as business production), our overall findings could imply that the program tended to crowd out new loans when the stimulus was on consumer durables and there was no impact for subsidies to productive business durable goods.

Table 4: New loan probability

	All			Passenger Cars			Trucks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	All	Unsecure	Secure	All	Unsecure	Secure	All	Unsecure	Secure
Treatment x r(-3)	0.010	0.010	-0.001	0.031***	0.029***	0.001	-0.018	-0.015	-0.003
	(0.007)	(0.006)	(0.002)	(0.006)	(0.005)	(0.003)	(0.015)	(0.013)	(0.007)
Treatment x r(-2)	-0.004	-0.001	-0.003	0.003	0.008	-0.004	-0.013	-0.013	-0.001
	(0.010)	(0.010)	(0.003)	(0.012)	(0.014)	(0.004)	(0.020)	(0.020)	(0.007)
Treatment x r(-1)	0.012	0.012	-0.001	0.017*	0.020*	-0.001	0.005	0.001	0.001
	(0.007)	(0.007)	(0.002)	(0.009)	(0.009)	(0.004)	(0.012)	(0.010)	(0.008)
Treatment x r(0)	-0.026	-0.025	-0.000	-0.028*	-0.023	-0.004	-0.022	-0.029	0.005
	(0.017)	(0.018)	(0.001)	(0.014)	(0.019)	(0.005)	(0.025)	(0.023)	(0.008)
Treatment x r(1)	-0.022**	-0.013	-0.009***	-0.028**	-0.015	-0.013*	-0.013	-0.012	-0.002
	(0.009)	(0.009)	(0.002)	(0.008)	(0.010)	(0.006)	(0.019)	(0.016)	(0.009)
Treatment x r(2)	-0.022*	-0.016	-0.009**	-0.028***	-0.015	-0.015**	-0.015	-0.017	0.000
	(0.010)	(0.009)	(0.004)	(0.008)	(0.010)	(0.005)	(0.019)	(0.013)	(0.011)
Treatment x r(3)	-0.021*	-0.009	-0.013***	-0.028**	-0.008	-0.018*	-0.012	-0.010	-0.005
	(0.010)	(0.013)	(0.002)	(0.010)	(0.015)	(0.009)	(0.020)	(0.017)	(0.009)
Treatment x r(4)	-0.030	-0.019	-0.013***	-0.041*	-0.027	-0.015*	-0.016	-0.010	-0.010
	(0.016)	(0.017)	(0.001)	(0.018)	(0.018)	(0.008)	(0.024)	(0.019)	(0.007)
Treatment x r(5)	-0.027	-0.014	-0.016***	-0.042**	-0.027	-0.019***	-0.011	-0.002	-0.011
	(0.017)	(0.017)	(0.004)	(0.015)	(0.016)	(0.004)	(0.027)	(0.021)	(0.012)
Treatment x r(6)	-0.016	-0.011	-0.008***	-0.024	-0.018	-0.005	-0.008	-0.005	-0.008
	(0.013)	(0.014)	(0.002)	(0.015)	(0.017)	(0.007)	(0.022)	(0.016)	(0.010)
Observations	169,271	169,271	169,271	88,170	88,170	88,170	81,101	81,101	81,101
Number of observations	18,829	18,829	18,829	9,571	9,571	9,571	9,258	9,258	9,258

Notes: The table reports the marginal effects of the treatment group at the quarter (m) relative to the purchase. Each regression includes both time- and individual-fixed effects. Standard errors are heteroscedasticity-robust and are clustered on cohort groups. *** = Significantly different from zero at the 1% level, ** = Significantly different from zero at the 5% level, * = Significantly different from zero at the 10% level.

When we allow for impact heterogeneity, our study shows that the crowd out impact of the program on new loan origination was smaller for those buyers who had prior access to commercial banks, i.e., the total impact of the program was less negative, confirming their higher ability to get loans from commercial banks. We also find that younger buyers experienced larger negative impact of the program on their likelihood of originating new loans when compared with the impact on older buyers. This finding suggests that there was higher likelihood that the program pushed the younger, relative to the older, to their debt limit, which consequently prevented them to take more loans. Finally, the crowd out effect of the program on secure loan origination for passenger car buyers tends to be less for urban than rural individuals. The regression results are presented in Tables C.1-C.3 in the appendix.

4. Empirical Analysis at the Postcode Level

So far our analyses have focused on the impacts of the first-time car buyer tax rebate program on individuals. However, the program could also generate impacts at a larger geographic level. In this section, we will explore possible general equilibrium impacts of the program by studying the debt at a more aggregate, namely, at the postcode level. In parallel to the analysis at the individual level, we examine two sets of outcome variables in this analysis: (1) the share of loan outstanding with delinquency status and (2) the growth of total loan outstanding. We also further consider the outcomes by loan types: all loans, auto loans, and non-auto loans.

4.1 Data

For the zip code level analysis, we begin with one million borrowers randomly sampled from the NCB database from December 2009 to December 2015. We then collapse that individual sample into the quarterly postcode panel. All loan outstanding variables are collapsed by summing over all individuals in each postcode. Table 5 presents summary statistics of outcome and exposure variables at the postcode level.

Table 5: Summary Statistics for Zip-level analysis

Panel A: Outcome Variables

	Pre			Post			Post-Pre
	Mean	SD	n	Mean	SD	n	
Loan growth (All)	0.070	0.092	10632	0.033	0.054	10632	-0.036*** (0.001)
Loan growth (Auto)	0.100	0.136	10632	0.038	0.075	10632	-0.063*** (0.002)
Loan growth (Non-Auto)	0.062	0.119	10632	0.033	0.070	10632	-0.029*** (0.001)
Delinquent loan Share (All)	0.067	0.043	11518	0.095	0.043	10632	0.028*** (0.001)
Delinquent loan Share (Auto)	0.070	0.053	11518	0.105	0.050	10632	0.036*** (0.001)
Delinquent loan Share (Non-Auto)	0.065	0.053	11518	0.089	0.052	10632	0.024*** (0.001)

Panel B: Exposure variable

	Mean	SD	n	10th	50th	90th
Ratio of first-car participants to all borrowers	0.068	0.021	886	0.042	0.066	0.094

Notes: Numbers in parentheses indicate standard error. *** = Significantly different from zero at the 1% level, ** = Significantly different from zero at the 5% level, * = Significantly different from zero at the 10% level.

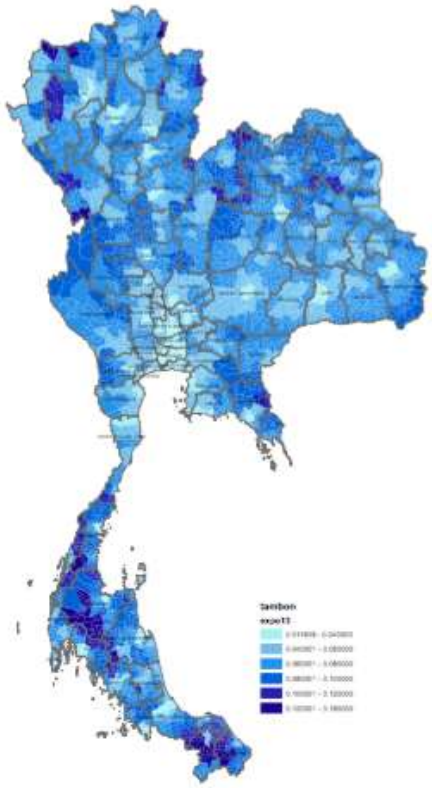
4.2 Empirical Strategy

Similar to previous analyses at the individual level, we use time-shifted difference-in-difference strategy to study the general equilibrium impacts of the program. Specifically, we use quarterly postcode-level data to estimate the following regression:

$$y_{it} = \alpha_1\, exposure_i + \alpha_2\, post_t + \beta_1 exposure_i \cdot post_t + province_year\ FE$$
$$+time\ FE + postcode\ FE + \varepsilon_{it}$$

where y_{it} is an outcome variable; $exposure_i$ is an exposure variable ranging from 0 to 1, $post_t$ is an indicator variable that equals 1 for years 2013-2015 and 0 otherwise. The exposure variable ($exposure_i$) is the ratio of the number of individual in the treatment group (as defined earlier in the previous section) to the total number of borrows in each postcode. Figure 2 illustrates the geographic distribution of exposure across postcodes.

Figure 2 Share of First-Car Scheme Exposure Across Postcodes



4.3 Empirical Results

Table 6 shows the regression results of the general equilibrium impacts of first-car buyer tax rebate scheme at the postcode level. The results are similar to what we find earlier at the individual level. Postcodes with higher exposure to the first-car tax rebate program tended to have higher share of loan outstanding with delinquency status (Panel A). These postcodes also experienced lower loan growth (Panel B). These findings are significant for all loans, auto loans only, and non-auto loans.

It is important to note that the general equilibrium effects could come from two sources. First, they could be just the aggregate of the program's impacts on loan delinquency and new loan origination at the individual level presented in the previous section. Second, the effects could come from the spillovers from program participants to non-participants living in the same local economy, i.e., the

same postcode. The channels of these spillovers could include possible lower spending of program participants due to their debt overhang, which was in turn a result from shifting future demand forward to the present while not financially ready for the debt services arising from the first-car loans. Note that these findings could be driven by (1) the effects from aggregation of program participants themselves, (2) the effects from spillovers from the participants to other borrowers, or both. We are in progress of separating the general equilibrium effects into these components. Finally, one may infer from Figure 2 that exposure to the first-car tax rebate was different between rural and urban areas, and out measure of exposure might just capture the difference in urbanization rather than the program itself. We provide in Appendix D that our overall results hold when we perform the analysis separately for urban postcodes and rural postcodes.

Table 6 Postcode regression analysis

	(1) All	(2) Auto	(3) Non-Auto
Panel A: Share of loan outstanding with delinquent status			
*exposure*post*	0.296***	0.309***	0.168**
	(0.076)	(0.105)	(0.072)
Observations	22,150	22,150	22,150
Number of zip	886	886	886
Fixed Effects		Postcode, Time	
Panel B: Growth of loan outstanding			
*exposure*post*	-0.354***	-0.488***	-0.165**
	(0.046)	(0.072)	(0.064)
Observations	21,264	21,264	21,264
Number of zip	886	886	886
Fixed Effects		Postcode, Time	

Notes: Robust standard errors are clustered at the province level. Numbers in parentheses indicate standard errors. *** = Significantly different from zero at the 1% level, ** = Significantly different from zero at the 5% level, * = Significantly different from zero at the 10% level.

6. Conclusion

This study uses account-level loan data from Thailand's National Credit Bureau to study the impacts of fiscal stimulus on household debt. Specifically, we analyze the impacts of the first-car buyer tax rebate scheme that was implemented in Thailand during 2011-13 on delinquency and new loan origination at the individual level as well as delinquency ratio and loan growth at the local economy (postcode) level. At the individual level, we find that delinquency probability for auto loans as well as other loans was significantly higher for the treatment group (individuals likely participating in the program), relative to that of the control group (those not eligible for the program). The overall new loan probability was also significantly lower for the treatment group. The effects were different between first-time passenger car buyers and first-time truck buyers, with the fall in new loan probability concentrated among passenger car purchases and no impact among truck buyers. This finding suggests that while the program crowded out new loans when the stimulus was on consumer durables, there was no impact for subsidies to productive business durables. At the more aggregate local economy level, we find that areas with higher exposure to the program experienced higher delinquency ratios of loan outstanding and lower loan growth rates. These results could be driven by both program participants themselves as well as spillover effects to other borrowers.

This paper shed light on the impacts of fiscal stimulus on the economy. It shows that there could be significant unintended adverse effects on program participants and local economies that could undermine the potential stimulus benefits. The study also highlights the important distinction between consumer and business durables. Finally, it illustrates how administrative micro-level data could help inform policymakers on the design of future fiscal stimulus programs.

Appendix A:

Table A.1: Basic difference-in-difference framework (Delinquency propensity)

Panel A: Delinquency (First car)

	A: Delinquency (First car)		
	Pre	**Post**	**Difference (Post-Pre)**
Control	N/A	0.038	N/A
		(0.001)	
Treatment	N/A	0.066	N/A
		(0.001)	
Difference (Treatment-Control)	N/A	0.027***	N/A
		(0.002)	

Panel B: Delinquency (Unsecure loans)

	B: Delinquency (All unsecure)		
	Pre	**Post**	**Difference (Post-Pre)**
Control	0.077***	0.119***	0.042***
	(0.003)	(0.002)	(0.004)
Treatment	0.080***	0.114***	0.035***
	(0.002)	(0.002)	(0.003)
Difference (Treatment-Control)	0.003	-0.005*	-0.008
	(0.003)	(0.003)	(0.005)

Panel C: Delinquency (Secure loans excluding first car)

	C: Delinquency (All secure)		
	Pre	**Post**	**Difference (Post-Pre)**
Control	0.044	0.075***	0.031***
	(0.004)***	(0.003)	(0.005)
Treatment	0.052	0.096***	0.044***
	(0.003)***	(0.003)	(0.004)
Difference (Treatment-Control)	0.009*	0.021***	0.013*
	(0.005)	(0.005)	0.007

Notes: Numbers in parentheses indicate standard error. *** = Significantly different from zero at the 1% level, ** = Significantly different from zero at the 5% level, * = Significantly different from zero at the 10% level.

Appendix B: Heterogeneous Impacts on Delinquency Probability

Table B.1 Delinquency probability allowing commercial bank access heterogeneity

	All			Passenger Cars			Trucks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	First car	Unsecure	Secure ex. First car	First car	Unsecure	Secure ex. First car	First car	Unsecure	Secure ex. First car
Treatment x r(-3)		-0.003	-0.026		0.005	-0.054		-0.011	0.019
		(0.005)	(0.013)		(0.006)	(0.036)		(0.014)	(0.019)
Treatment x r(-2)		0.002	-0.010		0.010	-0.026		-0.003	0.028
		(0.008)	(0.012)		(0.015)	(0.026)		(0.016)	(0.034)
Treatment x r(-1)		0.014	-0.017		0.022	-0.024		0.006	0.009
		(0.009)	(0.024)		(0.023)	(0.049)		(0.011)	(0.020)
Treatment x r(0)		0.012	-0.017**		0.020	-0.034		0.007	0.022
		(0.011)	(0.005)		(0.024)	(0.041)		(0.014)	(0.055)
Treatment x r(1)	0.016***	0.028***	-0.012	0.014***	0.045**	-0.047	0.020***	0.014	0.044***
	(0.002)	(0.005)	(0.029)	(0.002)	(0.017)	(0.052)	(0.005)	(0.021)	(0.010)
Treatment x r(2)	0.036***	0.033***	0.011	0.033***	0.047*	-0.027	0.042***	0.020	0.070***
	(0.005)	(0.008)	(0.027)	(0.006)	(0.019)	(0.061)	(0.005)	(0.037)	(0.014)
Treatment x r(3)	0.060***	0.030	-0.002	0.055***	0.039	-0.019	0.070**	0.023	0.042
	(0.008)	(0.018)	(0.042)	(0.004)	(0.031)	(0.057)	(0.019)	(0.014)	(0.027)
Treatment x r(4)	0.067***	0.032	0.016	0.057***	0.053	0.008	0.081**	0.008	0.054
	(0.009)	(0.026)	(0.049)	(0.004)	(0.040)	(0.062)	(0.024)	(0.010)	(0.028)
Treatment x r(5)	0.081***	0.036**	-0.022	0.063***	0.050***	-0.060*	0.104**	0.016	0.030
	(0.013)	(0.010)	(0.013)	(0.006)	(0.012)	(0.028)	(0.035)	(0.015)	(0.015)
Treatment x r(6)	0.081***	0.012	0.012	0.066***	0.024	-0.012	0.099**	-0.006	0.071
	(0.013)	(0.019)	(0.020)	(0.005)	(0.022)	(0.066)	(0.029)	(0.019)	(0.037)
Treatment x r(-3) x G		-0.001	0.019*		0.009	0.068*		-0.017	-0.056**
		(0.013)	(0.008)		(0.018)	(0.033)		(0.015)	(0.018)
Treatment x r(-2) x G		0.005	0.009		0.005	0.046*		0.006	-0.052*
		(0.008)	(0.007)		(0.021)	(0.019)		(0.021)	(0.024)
Treatment x r(-1) x G		-0.011	0.031		-0.010	0.058		-0.013	-0.020
		(0.006)	(0.026)		(0.034)	(0.051)		(0.032)	(0.010)

Table B.1 (cont.)

	All			Passenger Cars			Trucks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	First car	Unsecure	Secure ex. First car	First car	Unsecure	Secure ex. First car	First car	Unsecure	Secure ex. First car
Treatment x r(0) x G		-0.005	0.023		-0.010	0.053		-0.002	-0.035
		(0.008)	(0.019)		(0.023)	(0.044)		(0.038)	(0.048)
Treatment x r(1) x G	0.003	-0.013	0.041	-0.002	-0.022	0.099	0.009	-0.002	-0.045**
	(0.006)	(0.009)	(0.029)	(0.004)	(0.025)	(0.054)	(0.009)	(0.034)	(0.012)
Treatment x r(2) x G	0.008	-0.005	0.012	0.006	-0.014	0.064	0.012	0.012	-0.062**
	(0.009)	(0.019)	(0.017)	(0.013)	(0.032)	(0.046)	(0.009)	(0.042)	(0.020)
Treatment x r(3) x G	-0.014**	-0.001	0.036	-0.021**	0.003	0.053	-0.001	-0.008	0.000
	(0.005)	(0.026)	(0.036)	(0.008)	(0.035)	(0.050)	(0.010)	(0.027)	(0.038)
Treatment x r(4) x G	-0.012*	-0.009	0.004	-0.017*	-0.021	0.015	-0.003	0.009	-0.032
	(0.005)	(0.022)	(0.043)	(0.008)	(0.031)	(0.041)	(0.016)	(0.008)	(0.056)
Treatment x r(5) x G	-0.022*	-0.026	0.048***	-0.024	-0.025*	0.098***	-0.016	-0.016	-0.010
	(0.011)	(0.022)	(0.010)	(0.015)	(0.012)	(0.018)	(0.027)	(0.027)	(0.026)
Treatment x r(6) x G	-0.020	-0.004	0.006	-0.024	-0.002	0.026	-0.011	-0.006	-0.046
	(0.012)	(0.032)	(0.015)	(0.023)	(0.030)	(0.059)	(0.021)	(0.026)	(0.047)
Observations	58,134	73,831	22,699	32,381	45,034	13,837	25,753	28,797	8,862
Number of observations	8,358	7,234	2,846	4,656	4,328	1,687	3,702	2,906	1,159

Notes: The table reports the marginal effects of the treatment group at the quarter (m) relative to the purchase. *G* refers to the group dummy for individuals with prior access to commercial banks. Each regression includes both time- and individual-fixed effects. Standard errors are heteroscedasticity-robust and are clustered on cohort groups. *** = Significantly different from zero at the 1% level, ** = Significantly different from zero at the 5% level, * = Significantly different from zero at the 10% level.

Table B.2 Delinquency probability allowing age heterogeneity

	All			Passenger Cars			Trucks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	First car	Unsecure	Secure ex. First car	First car	Unsecure	Secure ex. First car	First car	Unsecure	Secure ex. First car
Treatment x r(-3)		-0.003	-0.009		0.004	-0.001		-0.012	-0.015
		(0.009)	(0.010)		(0.010)	(0.020)		(0.008)	(0.008)
Treatment x r(-2)		-0.002	-0.004		0.001	0.006		-0.002	-0.010
		(0.011)	(0.004)		(0.010)	(0.017)		(0.013)	(0.024)
Treatment x r(-1)		-0.009	0.008		0.001	0.012		-0.016	0.009
		(0.009)	(0.017)		(0.015)	(0.034)		(0.010)	(0.014)
Treatment x r(0)		-0.010	-0.013		-0.011	-0.005		-0.005	-0.013
		(0.006)	(0.008)		(0.015)	(0.028)		(0.013)	(0.025)
Treatment x r(1)	0.017***	0.010	0.021	0.014***	0.020	0.031	0.019***	0.003	0.022
	(0.001)	(0.007)	(0.022)	(0.004)	(0.016)	(0.032)	(0.004)	(0.016)	(0.016)
Treatment x r(2)	0.043***	0.031*	0.035	0.039***	0.039*	0.038	0.047***	0.026	0.048*
	(0.005)	(0.014)	(0.019)	(0.007)	(0.020)	(0.026)	(0.008)	(0.020)	(0.022)
Treatment x r(3)	0.054***	0.023	0.038**	0.044***	0.030	0.028	0.063***	0.019	0.064*
	(0.005)	(0.012)	(0.012)	(0.007)	(0.022)	(0.020)	(0.006)	(0.026)	(0.030)
Treatment x r(4)	0.065***	0.008	0.050**	0.054***	0.018	0.048	0.075***	0.000	0.066
	(0.004)	(0.015)	(0.014)	(0.008)	(0.027)	(0.028)	(0.005)	(0.020)	(0.039)
Treatment x r(5)	0.077***	0.017	0.036	0.062***	0.035	0.042	0.090***	0.000	0.034
	(0.007)	(0.017)	(0.018)	(0.009)	(0.020)	(0.024)	(0.008)	(0.032)	(0.037)
Treatment x r(6)	0.085***	0.010	0.049*	0.071***	0.027	0.054	0.097***	-0.007	0.047
	(0.010)	(0.016)	(0.023)	(0.013)	(0.021)	(0.039)	(0.009)	(0.028)	(0.044)
Treatment x r(-3) x G		0.005	0.012		0.009	-0.009		-0.012	0.045**
		(0.010)	(0.009)		(0.013)	(0.019)		(0.011)	(0.011)
Treatment x r(-2) x G		0.008	0.009		0.007	-0.026		0.010	0.054
		(0.012)	(0.009)		(0.012)	(0.014)		(0.014)	(0.030)
Treatment x r(-1) x G		0.020*	-0.021		0.004	-0.028		0.047***	-0.018
		(0.009)	(0.016)		(0.016)	(0.030)		(0.008)	(0.018)

Table B.2 (cont.)

	All			Passenger Cars			Trucks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	First car	Unsecure	Secure ex. First car	First car	Unsecure	Secure ex. First car	First car	Unsecure	Secure ex. First car
Treatment x r(0) x G		0.028**	0.002		0.021	-0.023		0.047***	0.031
		(0.008)	(0.013)		(0.016)	(0.023)		(0.004)	(0.036)
Treatment x r(1) x G	-0.003	0.004	-0.019	0.000	-0.009	-0.046	-0.002	0.029**	0.011
	(0.002)	(0.008)	(0.019)	(0.005)	(0.014)	(0.028)	(0.006)	(0.009)	(0.013)
Treatment x r(2) x G	-0.006	-0.028*	-0.017	-0.001	-0.037*	-0.060*	-0.008	-0.009	0.042**
	(0.005)	(0.014)	(0.014)	(0.007)	(0.017)	(0.023)	(0.009)	(0.014)	(0.015)
Treatment x r(3) x G	0.004	-0.004	0.000	0.007	-0.011	-0.019	0.008	0.006	0.045
	(0.005)	(0.010)	(0.015)	(0.006)	(0.014)	(0.021)	(0.010)	(0.017)	(0.030)
Treatment x r(4) x G	0.006	0.032**	-0.026	0.005	0.028	-0.053	0.015	0.032**	0.040
	(0.005)	(0.013)	(0.019)	(0.006)	(0.019)	(0.028)	(0.010)	(0.012)	(0.045)
Treatment x r(5) x G	0.003	0.009	-0.011	0.003	-0.008	-0.061*	0.012	0.033	0.083*
	(0.006)	(0.014)	(0.011)	(0.006)	(0.014)	(0.024)	(0.011)	(0.022)	(0.032)
Treatment x r(6) x G	0.004	0.010	-0.036	0.003	-0.007	-0.080*	0.014	0.028	0.055*
	(0.006)	(0.012)	(0.019)	(0.008)	(0.015)	(0.034)	(0.009)	(0.018)	(0.026)
Observations	131,264	97,328	28,434	66,705	59,674	16,925	64,559	37,654	11,509
Number of observations	18,829	10,674	3,817	9,571	6,436	2,195	9,258	4,238	1,622

Notes: The table reports the marginal effects of the treatment group at the quarter (m) relative to the purchase. *G* refers to the group dummy for individuals born in or after 1980. Each regression includes both time- and individual-fixed effects. Standard errors are heteroscedasticity-robust and are clustered on cohort groups. *** = Significantly different from zero at the 1% level, ** = Significantly different from zero at the 5% level, * = Significantly different from zero at the 10% level.

Table B.3 Delinquency probability allowing urbanization heterogeneity

	All			Passenger Cars			Trucks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	First car	Unsecure	Secure ex. First car	First car	Unsecure	Secure ex. First car	First car	Unsecure	Secure ex. First car
Treatment x r(-3)		-0.001	-0.001		0.014	-0.007		-0.018**	0.000
		(0.007)	(0.015)		(0.011)	(0.017)		(0.007)	(0.021)
Treatment x r(-2)		-0.000	0.011		0.018	0.020		-0.015	-0.001
		(0.010)	(0.014)		(0.011)	(0.011)		(0.009)	(0.031)
Treatment x r(-1)		-0.006	0.017		0.011	0.039		-0.020	-0.012
		(0.013)	(0.015)		(0.015)	(0.033)		(0.012)	(0.013)
Treatment x r(0)		-0.005	-0.023		0.003	-0.009		-0.013	-0.032
		(0.014)	(0.016)		(0.017)	(0.034)		(0.020)	(0.026)
Treatment x r(1)	0.016***	0.014	0.018	0.014**	0.031*	0.041	0.018***	-0.002	0.001
	(0.001)	(0.011)	(0.016)	(0.005)	(0.016)	(0.024)	(0.002)	(0.018)	(0.016)
Treatment x r(2)	0.042***	0.020*	0.049**	0.033***	0.030**	0.050	0.048***	0.011	0.063**
	(0.005)	(0.009)	(0.017)	(0.005)	(0.012)	(0.027)	(0.005)	(0.017)	(0.022)
Treatment x r(3)	0.053***	0.025	0.045**	0.042***	0.040**	0.034	0.063***	0.009	0.077*
	(0.009)	(0.015)	(0.014)	(0.010)	(0.016)	(0.019)	(0.009)	(0.022)	(0.031)
Treatment x r(4)	0.066***	0.028	0.073**	0.055***	0.049*	0.069*	0.076***	0.007	0.097*
	(0.009)	(0.019)	(0.019)	(0.011)	(0.020)	(0.031)	(0.010)	(0.022)	(0.038)
Treatment x r(5)	0.079***	0.031	0.053**	0.061***	0.048**	0.058*	0.093***	0.010	0.057*
	(0.013)	(0.022)	(0.017)	(0.013)	(0.017)	(0.024)	(0.015)	(0.032)	(0.028)
Treatment x r(6)	0.087***	0.026	0.072**	0.070***	0.050**	0.069	0.100***	-0.005	0.081***
	(0.011)	(0.020)	(0.019)	(0.012)	(0.019)	(0.037)	(0.014)	(0.026)	(0.020)
Treatment x r(-3) x G		-0.000	-0.005		-0.010	0.003		0.007	-0.007
		(0.011)	(0.018)		(0.010)	(0.019)		(0.021)	(0.013)
Treatment x r(-2) x G		0.003	-0.021		-0.023	-0.037**		0.035	0.010
		(0.015)	(0.016)		(0.013)	(0.012)		(0.027)	(0.016)
Treatment x r(-1) x G		0.010	-0.028*		-0.013	-0.063**		0.041	0.031
		(0.021)	(0.012)		(0.022)	(0.017)		(0.022)	(0.016)

Table B.3 (cont.)

	All			Passenger Cars			Trucks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	First car	Unsecure	Secure ex. First car	First car	Unsecure	Secure ex. First car	First car	Unsecure	Secure ex. First car
Treatment x r(0) x G		0.012	0.019		-0.009	-0.007		0.048	0.054*
		(0.022)	(0.025)		(0.012)	(0.042)		(0.036)	(0.024)
Treatment x r(1) x G	0.001	-0.002	-0.005	0.002	-0.026	-0.041*	0.002	0.033	0.046**
	(0.003)	(0.021)	(0.008)	(0.008)	(0.021)	(0.018)	(0.004)	(0.027)	(0.012)
Treatment x r(2) x G	-0.000	0.005	-0.031	0.011**	-0.010	-0.048	-0.009	0.030	-0.007
	(0.003)	(0.017)	(0.033)	(0.003)	(0.020)	(0.037)	(0.005)	(0.015)	(0.038)
Treatment x r(3) x G	0.007	-0.000	-0.011	0.012	-0.019	-0.018	0.008	0.026	0.001
	(0.010)	(0.021)	(0.022)	(0.012)	(0.015)	(0.023)	(0.010)	(0.034)	(0.023)
Treatment x r(4) x G	0.004	-0.006	-0.049	0.005	-0.023	-0.060	0.011	0.011	-0.036*
	(0.010)	(0.021)	(0.033)	(0.011)	(0.025)	(0.042)	(0.013)	(0.023)	(0.016)
Treatment x r(5) x G	0.001	-0.010	-0.030	0.006	-0.021	-0.053	0.003	0.009	0.003
	(0.013)	(0.024)	(0.024)	(0.013)	(0.023)	(0.029)	(0.016)	(0.029)	(0.019)
Treatment x r(6) x G	0.001	-0.011	-0.054	0.006	-0.034	-0.064***	0.004	0.021	-0.033
	(0.007)	(0.020)	(0.028)	(0.009)	(0.022)	(0.016)	(0.008)	(0.014)	(0.040)
Observations	131,264	97,328	28,434	66,705	59,674	16,925	64,559	37,654	11,509
Number of observations	18,829	10,674	3,817	9,571	6,436	2,195	9,258	4,238	1,622

Notes: The table reports the marginal effects of the treatment group at the quarter (m) relative to the purchase. *G* refers to the group dummy for individuals living urban areas. Each regression includes both time- and individual-fixed effects. Standard errors are heteroscedasticity-robust and are clustered on cohort groups. *** = Significantly different from zero at the 1% level, ** = Significantly different from zero at the 5% level, * = Significantly different from zero at the 10% level.

Appendix C: Heterogeneous Impacts on New Loan Probability

Table C.1 New loan probability allowing commercial bank access heterogeneity

	All			Passenger Cars			Trucks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	All	Unsecure	Secure	All	Unsecure	Secure	All	Unsecure	Secure
Treatment x r(-3)	-0.012*	-0.008*	-0.005*	-0.009	-0.001	-0.009	-0.016	-0.016	-0.002
	(0.006)	(0.004)	(0.002)	(0.005)	(0.004)	(0.005)	(0.011)	(0.009)	(0.004)
Treatment x r(-2)	-0.031***	-0.028**	-0.005	-0.036***	-0.035***	-0.001	-0.029	-0.023	-0.010*
	(0.008)	(0.010)	(0.004)	(0.005)	(0.005)	(0.004)	(0.017)	(0.020)	(0.005)
Treatment x r(-1)	-0.007	-0.002	-0.009***	-0.019	-0.013	-0.011	0.006	0.012	-0.007
	(0.019)	(0.019)	(0.003)	(0.014)	(0.016)	(0.006)	(0.026)	(0.024)	(0.006)
Treatment x r(0)	-0.023*	-0.021	-0.003	-0.034***	-0.031**	-0.004	-0.012	-0.013	-0.001
	(0.009)	(0.013)	(0.004)	(0.003)	(0.008)	(0.007)	(0.017)	(0.021)	(0.003)
Treatment x r(1)	-0.029***	-0.020**	-0.011**	-0.045*	-0.037	-0.012	-0.007	-0.000	-0.007*
	(0.005)	(0.007)	(0.004)	(0.020)	(0.020)	(0.008)	(0.010)	(0.014)	(0.003)
Treatment x r(2)	-0.016	-0.009	-0.011*	-0.013	-0.007	-0.007	-0.015	-0.008	-0.013
	(0.010)	(0.012)	(0.005)	(0.019)	(0.015)	(0.006)	(0.026)	(0.024)	(0.016)
Treatment x r(3)	-0.006	0.003	-0.013**	-0.024**	-0.008	-0.017	0.017	0.016	-0.007
	(0.006)	(0.007)	(0.005)	(0.009)	(0.010)	(0.013)	(0.019)	(0.013)	(0.012)
Treatment x r(4)	-0.027	-0.021	-0.010	-0.052	-0.039	-0.016***	0.003	0.000	-0.001
	(0.027)	(0.024)	(0.005)	(0.031)	(0.029)	(0.004)	(0.028)	(0.026)	(0.009)
Treatment x r(5)	-0.027	-0.018	-0.016*	-0.059**	-0.048**	-0.023**	0.007	0.013	-0.007
	(0.023)	(0.021)	(0.008)	(0.021)	(0.018)	(0.006)	(0.034)	(0.027)	(0.017)
Treatment x r(6)	-0.015**	-0.008	-0.012*	-0.041**	-0.032**	-0.008	0.012	0.017	-0.014
	(0.005)	(0.004)	(0.005)	(0.016)	(0.012)	(0.007)	(0.010)	(0.012)	(0.007)
Treatment x r(-3) x G	0.039**	0.034**	0.007	0.058***	0.044***	0.014	0.018	0.021	-0.000
	(0.011)	(0.011)	(0.005)	(0.013)	(0.007)	(0.009)	(0.012)	(0.019)	(0.012)
Treatment x r(-2) x G	0.027***	0.036***	-0.007	0.034*	0.046**	-0.011	0.026	0.029	-0.001
	(0.005)	(0.005)	(0.006)	(0.014)	(0.015)	(0.007)	(0.019)	(0.018)	(0.009)
Treatment x r(-1) x G	0.006	0.009	0.001	0.022*	0.022*	0.006	-0.006	-0.002	-0.003
	(0.013)	(0.010)	(0.004)	(0.010)	(0.010)	(0.006)	(0.022)	(0.022)	(0.004)

Table C.1 (cont.)

	All			Passenger Cars			Trucks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	All	Unsecure	Secure	All	Unsecure	Secure	All	Unsecure	Secure
Treatment x r(0) x G	0.019	0.030**	-0.013*	0.047**	0.057***	-0.011	-0.006	0.004	-0.015
	(0.011)	(0.009)	(0.007)	(0.015)	(0.013)	(0.007)	(0.023)	(0.026)	(0.010)
Treatment x r(1) x G	0.015	0.016	0.003	0.031	0.036	-0.001	0.002	-0.003	0.009
	(0.016)	(0.012)	(0.010)	(0.023)	(0.026)	(0.009)	(0.022)	(0.014)	(0.014)
Treatment x r(2) x G	-0.012	-0.010	-0.005	-0.001	0.003	-0.012*	-0.027*	-0.026	0.004
	(0.009)	(0.009)	(0.007)	(0.015)	(0.019)	(0.006)	(0.013)	(0.015)	(0.010)
Treatment x r(3) x G	-0.004	0.006	-0.004	0.011	0.022	-0.004	-0.020	-0.013	-0.002
	(0.011)	(0.010)	(0.006)	(0.011)	(0.015)	(0.014)	(0.025)	(0.024)	(0.004)
Treatment x r(4) x G	-0.006	0.001	-0.006	0.005	0.013	-0.006	-0.014	-0.011	-0.003
	(0.008)	(0.009)	(0.004)	(0.021)	(0.021)	(0.006)	(0.016)	(0.018)	(0.005)
Treatment x r(5) x G	0.005	0.015	-0.006	0.026	0.034	0.005	-0.014	-0.002	-0.017
	(0.008)	(0.009)	(0.004)	(0.021)	(0.020)	(0.010)	(0.015)	(0.011)	(0.011)
Treatment x r(6) x G	0.011	0.018	-0.004	0.031	0.041	-0.007	-0.004	-0.000	-0.003
	(0.024)	(0.020)	(0.011)	(0.025)	(0.022)	(0.009)	(0.033)	(0.023)	(0.019)
Observations	90,627	90,627	90,627	50,617	50,617	50,617	40,010	40,010	40,010
Number of observations	8,358	8,358	8,358	4,656	4,656	4,656	3,702	3,702	3,702

Notes: The table reports the marginal effects of the treatment group at the quarter (m) relative to the purchase. *G* refers to the group dummy for individuals with prior access to commercial banks. Each regression includes both time- and individual-fixed effects. Standard errors are heteroscedasticity-robust and are clustered on cohort groups. *** = Significantly different from zero at the 1% level, ** = Significantly different from zero at the 5% level, * = Significantly different from zero at the 10% level.

Table C.2 New loan probability allowing age heterogeneity

	All			Passenger Cars			Trucks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	All	Unsecure	Secure	All	Unsecure	Secure	All	Unsecure	Secure
Treatment x r(-3)	0.016***	0.016***	0.001	0.035***	0.034***	0.001	-0.005**	-0.004	0.001
	(0.004)	(0.003)	(0.003)	(0.007)	(0.002)	(0.005)	(0.001)	(0.007)	(0.008)
Treatment x r(-2)	-0.005	0.001	-0.006**	-0.009	-0.001	-0.007	0.001	0.005	-0.004
	(0.015)	(0.016)	(0.002)	(0.013)	(0.018)	(0.006)	(0.021)	(0.016)	(0.009)
Treatment x r(-1)	0.018***	0.020***	-0.002	0.024**	0.032***	-0.005	0.014*	0.009	0.002
	(0.004)	(0.004)	(0.002)	(0.008)	(0.004)	(0.006)	(0.006)	(0.008)	(0.010)
Treatment x r(0)	-0.007	-0.005	-0.003	-0.006	0.005	-0.011	-0.003	-0.009	0.006
	(0.011)	(0.011)	(0.002)	(0.014)	(0.019)	(0.006)	(0.017)	(0.014)	(0.009)
Treatment x r(1)	-0.013*	-0.004	-0.010**	-0.027*	-0.009	-0.018*	0.003	0.002	-0.000
	(0.006)	(0.007)	(0.003)	(0.013)	(0.016)	(0.009)	(0.012)	(0.011)	(0.011)
Treatment x r(2)	-0.016**	-0.010*	-0.009	-0.029***	-0.011	-0.022**	0.000	-0.006	0.004
	(0.007)	(0.005)	(0.005)	(0.006)	(0.010)	(0.008)	(0.016)	(0.009)	(0.012)
Treatment x r(3)	-0.016**	0.001	-0.018***	-0.028**	0.004	-0.032**	-0.000	0.003	-0.006
	(0.006)	(0.007)	(0.004)	(0.011)	(0.006)	(0.009)	(0.016)	(0.015)	(0.011)
Treatment x r(4)	-0.017**	-0.004	-0.018**	-0.035*	-0.015	-0.023	0.003	0.009	-0.011
	(0.006)	(0.006)	(0.006)	(0.015)	(0.009)	(0.013)	(0.017)	(0.014)	(0.008)
Treatment x r(5)	-0.014*	-0.000	-0.017**	-0.041**	-0.021	-0.025***	0.013	0.020	-0.008
	(0.007)	(0.005)	(0.005)	(0.012)	(0.011)	(0.007)	(0.016)	(0.011)	(0.013)
Treatment x r(6)	-0.013	-0.003	-0.013**	-0.026**	-0.013	-0.013	0.005	0.009	-0.010
	(0.011)	(0.011)	(0.005)	(0.011)	(0.010)	(0.011)	(0.022)	(0.017)	(0.011)
Treatment x r(-3) x G	-0.016	-0.015	-0.005	-0.008	-0.011	0.001	-0.048**	-0.038**	-0.015
	(0.014)	(0.011)	(0.004)	(0.015)	(0.011)	(0.006)	(0.014)	(0.011)	(0.009)
Treatment x r(-2) x G	-0.002	-0.009	0.008**	0.023	0.018	0.008	-0.050*	-0.063**	0.010
	(0.017)	(0.018)	(0.003)	(0.017)	(0.020)	(0.007)	(0.023)	(0.019)	(0.011)
Treatment x r(-1) x G	-0.019*	-0.023***	0.004	-0.021	-0.032***	0.008	-0.032***	-0.028***	-0.001
	(0.009)	(0.006)	(0.003)	(0.013)	(0.008)	(0.007)	(0.006)	(0.006)	(0.008)

Table C.2 (cont.)

	All			Passenger Cars			Trucks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	All	Unsecure	Secure	All	Unsecure	Secure	All	Unsecure	Secure
Treatment x r(0) x G	-0.045***	-0.051***	0.008***	-0.042**	-0.056**	0.017**	-0.065***	-0.064***	-0.002
	(0.011)	(0.011)	(0.001)	(0.016)	(0.020)	(0.006)	(0.014)	(0.013)	(0.007)
Treatment x r(1) x G	-0.018*	-0.022*	0.005	0.004	-0.011	0.015	-0.055***	-0.048***	-0.004
	(0.009)	(0.010)	(0.003)	(0.016)	(0.017)	(0.008)	(0.010)	(0.011)	(0.009)
Treatment x r(2) x G	-0.013	-0.014	0.002	0.007	-0.007	0.016*	-0.050***	-0.040**	-0.012
	(0.009)	(0.011)	(0.004)	(0.012)	(0.015)	(0.008)	(0.012)	(0.012)	(0.010)
Treatment x r(3) x G	-0.010	-0.025**	0.017***	0.003	-0.026**	0.032***	-0.039**	-0.042*	0.002
	(0.009)	(0.011)	(0.003)	(0.011)	(0.009)	(0.008)	(0.014)	(0.017)	(0.005)
Treatment x r(4) x G	-0.032***	-0.039***	0.011	-0.013	-0.025	0.018	-0.066***	-0.067***	0.003
	(0.008)	(0.009)	(0.007)	(0.019)	(0.016)	(0.011)	(0.012)	(0.012)	(0.005)
Treatment x r(5) x G	-0.032**	-0.035**	0.003	-0.001	-0.012	0.014*	-0.081***	-0.073***	-0.012
	(0.011)	(0.013)	(0.002)	(0.020)	(0.019)	(0.007)	(0.009)	(0.013)	(0.011)
Treatment x r(6) x G	-0.010	-0.020	0.015***	0.006	-0.011	0.019	-0.047**	-0.047**	0.004
	(0.011)	(0.014)	(0.004)	(0.017)	(0.016)	(0.011)	(0.014)	(0.015)	(0.005)
Observations	169,271	169,271	169,271	88,170	88,170	88,170	81,101	81,101	81,101
Number of observations	18,829	18,829	18,829	9,571	9,571	9,571	9,258	9,258	9,258

Notes: The table reports the marginal effects of the treatment group at the quarter (m) relative to the purchase. *G* refers to the group dummy for individuals born in or after 1980. Each regression includes both time- and individual-fixed effects. Standard errors are heteroscedasticity-robust and are clustered on cohort groups. *** = Significantly different from zero at the 1% level, ** = Significantly different from zero at the 5% level, * = Significantly different from zero at the 10% level.

Table C.3 New loan probability allowing urbanization heterogeneity

	All			Passenger Cars			Trucks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	All	Unsecure	Secure	All	Unsecure	Secure	All	Unsecure	Secure
Treatment x r(-3)	-0.001	0.006	-0.006	0.011	0.017	-0.003	-0.014	-0.007	-0.007
	(0.018)	(0.015)	(0.007)	(0.018)	(0.019)	(0.007)	(0.029)	(0.024)	(0.010)
Treatment x r(-2)	-0.010	-0.006	-0.005	-0.004	0.005	-0.008	-0.016	-0.017	0.001
	(0.019)	(0.019)	(0.002)	(0.023)	(0.026)	(0.007)	(0.027)	(0.028)	(0.005)
Treatment x r(-1)	0.012	0.012	-0.001	0.012	0.018	-0.004	0.012	0.006	0.004
	(0.009)	(0.012)	(0.003)	(0.010)	(0.018)	(0.011)	(0.018)	(0.020)	(0.005)
Treatment x r(0)	-0.035**	-0.034*	-0.001	-0.046**	-0.034	-0.012	-0.021	-0.032	0.011
	(0.014)	(0.015)	(0.001)	(0.017)	(0.022)	(0.007)	(0.023)	(0.021)	(0.006)
Treatment x r(1)	-0.032*	-0.022	-0.011***	-0.050**	-0.031	-0.021**	-0.012	-0.013	0.001
	(0.015)	(0.014)	(0.002)	(0.019)	(0.024)	(0.007)	(0.020)	(0.019)	(0.007)
Treatment x r(2)	-0.030*	-0.018	-0.013**	-0.054***	-0.021	-0.034**	-0.006	-0.012	0.007
	(0.013)	(0.013)	(0.005)	(0.012)	(0.019)	(0.010)	(0.023)	(0.017)	(0.010)
Treatment x r(3)	-0.026	-0.014	-0.014***	-0.048*	-0.017	-0.031**	-0.005	-0.009	0.002
	(0.018)	(0.021)	(0.003)	(0.021)	(0.029)	(0.009)	(0.021)	(0.021)	(0.007)
Treatment x r(4)	-0.032	-0.021	-0.016***	-0.046*	-0.025	-0.023**	-0.015	-0.014	-0.006
	(0.022)	(0.022)	(0.002)	(0.023)	(0.026)	(0.008)	(0.026)	(0.021)	(0.009)
Treatment x r(5)	-0.036	-0.020	-0.018**	-0.056*	-0.031	-0.029***	-0.015	-0.010	-0.006
	(0.025)	(0.024)	(0.006)	(0.026)	(0.028)	(0.004)	(0.027)	(0.022)	(0.011)
Treatment x r(6)	-0.026	-0.014	-0.014***	-0.046*	-0.025	-0.020**	-0.005	-0.003	-0.005
	(0.022)	(0.023)	(0.001)	(0.023)	(0.028)	(0.008)	(0.026)	(0.021)	(0.008)
Treatment x r(-3) x G	0.021	0.009	0.009	0.034	0.020	0.008	-0.011	-0.019	0.008
	(0.028)	(0.024)	(0.010)	(0.032)	(0.030)	(0.010)	(0.033)	(0.029)	(0.015)
Treatment x r(-2) x G	0.012	0.010	0.003	0.012	0.006	0.008	0.006	0.008	-0.005
	(0.024)	(0.022)	(0.006)	(0.027)	(0.027)	(0.008)	(0.024)	(0.024)	(0.008)
Treatment x r(-1) x G	0.001	0.001	0.001	0.009	0.005	0.005	-0.017	-0.012	-0.006
	(0.007)	(0.011)	(0.009)	(0.013)	(0.021)	(0.013)	(0.023)	(0.025)	(0.008)

Table C.3 (cont.)

	All			Passenger Cars			Trucks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	All	Unsecure	Secure	All	Unsecure	Secure	All	Unsecure	Secure
Treatment x r(0) x G	0.018	0.017	0.001	0.031	0.019	0.013*	-0.004	0.008	-0.014
	(0.010)	(0.010)	(0.004)	(0.022)	(0.022)	(0.006)	(0.017)	(0.015)	(0.008)
Treatment x r(1) x G	0.020	0.018	0.005**	0.039	0.029	0.014**	-0.002	0.003	-0.005
	(0.011)	(0.011)	(0.002)	(0.025)	(0.024)	(0.004)	(0.023)	(0.025)	(0.005)
Treatment x r(2) x G	0.015	0.003	0.009	0.046*	0.011	0.033**	-0.021	-0.013	-0.015
	(0.013)	(0.012)	(0.008)	(0.020)	(0.020)	(0.010)	(0.024)	(0.019)	(0.011)
Treatment x r(3) x G	0.011	0.009	0.003	0.035	0.016	0.023***	-0.016	-0.003	-0.017
	(0.019)	(0.020)	(0.003)	(0.028)	(0.028)	(0.004)	(0.012)	(0.018)	(0.009)
Treatment x r(4) x G	0.004	0.004	0.004	0.007	-0.004	0.013***	-0.001	0.011	-0.008
	(0.015)	(0.014)	(0.004)	(0.017)	(0.016)	(0.003)	(0.021)	(0.026)	(0.006)
Treatment x r(5) x G	0.018	0.013	0.004	0.024	0.007	0.017*	0.012	0.019	-0.012**
	(0.019)	(0.016)	(0.004)	(0.029)	(0.026)	(0.008)	(0.022)	(0.025)	(0.004)
Treatment x r(6) x G	0.019	0.006	0.012***	0.040	0.014	0.027***	-0.005	-0.004	-0.007
	(0.019)	(0.020)	(0.003)	(0.027)	(0.025)	(0.006)	(0.021)	(0.025)	(0.010)
Observations	169,271	169,271	169,271	88,170	88,170	88,170	81,101	81,101	81,101
Number of observations	18,829	18,829	18,829	9,571	9,571	9,571	9,258	9,258	9,258

Notes: The table reports the marginal effects of the treatment group at the quarter (m) relative to the purchase. *G* refers to the group dummy for individuals living in urban areas. Each regression includes both time- and individual-fixed effects. Standard errors are heteroscedasticity-robust and are clustered on cohort groups. *** = Significantly different from zero at the 1% level, ** = Significantly different from zero at the 5% level, * = Significantly different from zero at the 10% level.

Appendix D: Robustness for the Analysis at the Postcode Level

Table D.1: Postcode regression analysis

Panel A: Within-urban regression (Top 50% in term of urbanization degree)

VARIABLES	(1) All	(2) Auto	(3) Non-Auto
Panel A: Growth of loan outstanding			
*exposure*post*	-0.384***	-0.691***	-0.111
	(0.053)	(0.097)	(0.082)
Observations	10,344	10,344	10,344
Number of zip	431	431	431
Fixed Effects		Postcode, Time	
Panel B: Share of loan outstanding with delinquent status			
*exposure*post*	0.363***	0.245*	0.310***
	(0.087)	(0.124)	(0.098)
Observations	10,775	10,775	10,775
Number of zip	431	431	431
Fixed Effects		Postcode, Time	

Panel B: Within-rural regression (Bottom 50% in term of urbanization degree)

VARIABLES	(1) All	(2) Auto	(3) Non-Auto
Panel A: Growth of loan outstanding			
*exposure*post*	-0.346***	-0.397***	-0.192*
	(0.084)	(0.108)	(0.104)
Observations	10,920	10,920	10,920
Number of zip	455	455	455
Fixed Effects		Postcode, Time	
Panel B: Share of loan outstanding with delinquent status			
*exposure*post*	0.232*	0.403***	-0.012
	(0.123)	(0.145)	(0.123)
Observations	11,375	11,375	11,375
Number of zip	455	455	455
Fixed Effects		Postcode, Time	

Notes: Robust standard errors are clustered at the province level. Numbers in parentheses indicate standard errors. *** = Significantly different from zero at the 1% level, ** = Significantly different from zero at the 5% level, * = Significantly different from zero at the 10% level.